

Data Analysis

Student Core Reader v1

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Notation Guide

This guide is a compact reference for the symbols used across the regression portion of the Core Reader. The full reference build contains additional notation detail and Math Foundations source-map tables; package-specific code lives in the separate Coding and Software Cookbook companion.

Object Types

Notation	Object type	Core meaning
x, y, n, p	scalars	Single values, counts, or indices.
X, Y, ϵ	random quantities	Population/model-level random variables.
$\mathbf{x}, \mathbf{y}, \mathbf{e}$	observed vectors	Sample-level predictor, response, and residual vectors.
$\mathbf{X}, \mathbf{H}, \mathbf{I}_n$	matrices	Design, projection, and identity matrices.
$\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$	coefficient vectors	True and estimated regression coefficients.

Data and Regression Objects

For one observation, $\mathbf{x} = (x_1, \dots, x_p)^T$ is the predictor vector and Y is the response. The general supervised-learning model is

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is unexplained error.

For a sample of size n , the observed response vector is

$$\mathbf{y} = (y_1, \dots, y_n)^T.$$

In regression, the design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ usually includes an intercept column, and

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}.$$

When $\mathbf{X}$ has full column rank,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad \hat{\mathbf{y}} = \mathbf{H} \mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.$$

The matrix $\mathbf{H}$ is the *hat matrix*. It turns observed responses into fitted responses.

Errors, Residuals, and Assumptions

The error vector $\boldsymbol{\epsilon}$ belongs to the data-generating model and is not observed. The residual vector $\mathbf{e}$ is computed after fitting the model. In Chapter 10, the text temporarily writes $\mathbf{Y}$ for the random response vector under the fixed-design model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.$$

Most other chapters use $\mathbf{y}$ for the observed response vector from the data set in hand.

The regression chapters use the same three-step assumption ladder:

1. **Mean-zero errors:** $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$.
2. **Constant variance and uncorrelated errors:** $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$.
3. **Gaussian errors:** $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$.

Stronger assumptions support stronger inferential claims.

Fit and Diagnostic Quantities

Notation	Meaning
RSS	Residual sum of squares, $\sum_i (y_i - \hat{y}_i)^2 = \ \mathbf{e}\ ^2$.
TSS	Total sum of squares relative to the mean-only baseline.
R^2	Fraction of response variation explained relative to that baseline.
RSE	Estimated residual standard deviation.
h_{ii}	Leverage of observation i , the i th diagonal entry of $\mathbf{H}$.

Linear Models

Early chapters use “linear” in the familiar geometric sense: a line, plane, or hyperplane in the displayed predictors. Chapter 11 sharpens the definition: for least squares, the key requirement is linearity in the unknown coefficients, not straightness in the original variables.

Chapter 1

Data Analysis as Statistical Learning

1.1 Why This Course Begins with Statistical Learning

Data analysis is not just the act of computing summaries or fitting models. It is the disciplined process of using data to learn something about a system, make predictions about future observations, and communicate conclusions in a way that can be checked, reproduced, and defended. In this course, we use the phrase *statistical learning* to emphasize that our models are tools for learning from data rather than formulas that are true simply because we wrote them down.

This course serves as a bridge. It connects the probability, statistics, and mathematical tools developed in earlier operations analysis courses to the modeling and machine-learning methods that appear later. The immediate goal is to build the statistical modeling intuition needed for regression, exploratory data analysis, model validation, and diagnostics. The longer-term goal is to prepare you to use more advanced methods without treating them as black boxes.

A recurring theme is the statement, often summarized as “all models are wrong.” The point is not that models are useless. The point is that a model is a deliberate simplification. It keeps the structure we think is useful and ignores or absorbs the rest as noise, approximation error, measurement error, or unexplained variation. A good analyst understands both sides: what the model is trying to capture and what it is failing to capture.

1.2 What Is Data?

A data set is a collection of measurements, records, or observations. Each observation corresponds to one unit of analysis: a person, ship, aircraft, market, mission, experiment, time step, or other object being studied. Each variable records one attribute of that observation.

In statistical modeling, we usually distinguish between *input variables* and an *output variable*. Input variables are the measured quantities we use to explain or predict something. They are also called *predictors*, *features*, *covariates*, or *independent variables*. The output variable is the quantity we are trying to explain or predict. It is also called the *response* or *dependent variable*.

For example, suppose we want to model a service member’s physical fitness test score. Possible inputs might include age, training frequency, prior score, sleep, and injury status. The output could be the next test score. In a logistics setting, inputs might be demand history, location, distance,

and inventory status, while the output could be delivery time or stockout risk. In a classification problem, inputs might be sensor readings and the output might be whether a track is classified as friendly, hostile, or unknown.

We collect the input variables for a single observation into a vector. For an observation with p inputs, write

$$\mathbf{x} = (x_1, x_2, \dots, x_p)^T \in \mathbb{R}^p.$$

Here p is the number of input variables, and x_j is the value of the j th input for this observation. The corresponding output is denoted by Y or, for a realized observed value, by y .

Math Foundations Connection. Math Foundations treats a vector as an ordered collection of quantities that belong to the same object, not merely as an arrow in physical space. That is exactly how we use $\mathbf{x}$ here. The vector $\mathbf{x}$ is one observation's collection of input measurements. Its entries may be physical quantities, demographic attributes, encoded categories, time summaries, or engineered features. The geometric interpretation of vectors will become useful later, but the first interpretation is simpler: a feature vector is a structured record of measured information. See the Math Foundations textbook, Chapter 2.

1.3 A First Statistical Model

The central modeling idea in this chapter is that an output variable Y is related to input variables $\mathbf{X} = (X_1, X_2, \dots, X_p)^T$ through an unknown relationship plus random error:

$$Y = f(\mathbf{X}) + \epsilon. \tag{1.1}$$

The function f represents the systematic part of the relationship between the inputs and the output. The random variable ϵ represents the part of Y not explained by $f(\mathbf{X})$.

A simple example with two predictors is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \tag{1.2}$$

In this model, β_0 is an intercept, β_1 and β_2 are coefficients, and ϵ is the unobserved error. The inputs x_1 and x_2 are observed. The output Y is observed after the fact. The error term ϵ is not directly observed; it is the difference between the observed output and the systematic part of the model.

Equation (1.1) is intentionally general. The function f might be linear, nonlinear, simple, complicated, interpretable, or opaque. In this course, we often begin with linear models because they are mathematically transparent and remain important for inference. But the broader statistical learning framework is not limited to linear models.

We usually impose basic assumptions on the error term. A common starting point is

$$\mathbb{E}[\epsilon] = 0, \tag{1.3}$$

$$\text{Var}(\epsilon) > 0, \tag{1.4}$$

$$\epsilon \text{ is independent of } \mathbf{X}. \tag{1.5}$$

The mean-zero assumption says that the error does not systematically push the output upward or downward once the systematic relationship has been accounted for. The positive-variance assumption

says that there is genuine unexplained variability. The independence assumption says that the leftover error is not still carrying predictable information from the inputs. These assumptions are not decorations. They are part of the model’s claim about the world, and later chapters will show how violations of assumptions affect inference, prediction, and diagnostics.

Since f is unknown, we estimate it using data. We denote the estimated function by $\hat{f}$. Given a new input vector $\mathbf{x}$, the corresponding predicted output is

$$\hat{Y} = \hat{f}(\mathbf{x}). \quad (1.6)$$

The hat notation signals that the quantity is estimated or predicted rather than directly observed or known.

Math Foundations Connection. When the model is linear in the input variables, the systematic part can be written compactly as

$$f(\mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p = \beta_0 + \boldsymbol{\beta}^T \mathbf{x},$$

where $\boldsymbol{\beta} = (\beta_1, \dots, \beta_p)^T$. The term $\boldsymbol{\beta}^T \mathbf{x}$ is an inner product. Equivalently, it is a linear combination of the entries of $\mathbf{x}$, with coefficients $\beta_1, \dots, \beta_p$. This is why linear algebra becomes the natural language of regression: the model combines measured inputs by weighting them and adding them together. See the Math Foundations textbook, Chapters 2, 3, and 5.

1.4 Inference and Prediction

There are two major reasons to estimate f : inference and prediction. They are related, but they are not the same task.

1.4.1 Inference

Inference is about understanding the relationship between the inputs and the output. In the linear model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon,$$

inference focuses on questions such as:

- What is the meaning of β_1 and β_2 ?
- Are these coefficients positive, negative, or plausibly zero?
- Which predictors are most relevant to Y ?
- How much does the output change when one input changes and the others are held fixed?
- What assumptions are required before we trust those conclusions?

Inference is about learning structure. If a commander asks why readiness changed, which drivers matter, or whether a particular intervention had an effect, then the analysis is inferential. Accuracy still matters, but the model must also be interpretable enough to support an explanation.

1.4.2 Prediction

Prediction is about estimating the output for new or future inputs. Given a new vector $\mathbf{x}_{\text{new}}$, we use the fitted model to compute

$$\hat{Y}_{\text{new}} = \hat{f}(\mathbf{x}_{\text{new}}).$$

Prediction focuses on questions such as:

- Given these inputs, what output should we expect?
- How accurate is the prediction likely to be?
- How much uncertainty remains even after we use the model?
- How well will the model perform on cases it has not yet seen?

Prediction is about learning what is likely to happen next. If an analyst is building a model to forecast demand, estimate casualty risk, classify a sensor return, or predict equipment failure, then the model may be judged primarily by its predictive performance.

1.4.3 The Same Model Can Serve Both Goals

The same mathematical model can sometimes support both inference and prediction, but the priorities differ. A highly interpretable model may be preferred for inference even if a more flexible model predicts slightly better. Conversely, in a pure prediction task, a less interpretable model may be acceptable if it performs substantially better on new data. A defensible analysis states which goal is primary and evaluates the model accordingly.

1.5 Regression and Classification

A first way to categorize statistical learning problems is by the type of output variable.

In a *regression* problem, the output is numerical or quantitative. Examples include predicting fuel consumption, time to complete a task, exam score, cost, weight, or sales. A regression model returns a number. A simple linear regression model has the form

$$\hat{y} = wx + b, \tag{1.7}$$

where w is a slope or weight and b is an intercept or bias term.

In a *classification* problem, the output is a category or class label. Examples include predicting pass/fail, disease/no disease, hostile/friendly/unknown, or low/medium/high risk. A classifier may return a class directly, but many classifiers first estimate probabilities for each class and then choose the most likely class.

For binary classification, where $Y \in \{0, 1\}$, logistic regression models the probability of class 1 as

$$\Pr(Y = 1 \mid \mathbf{x}) = \sigma(w_1x_1 + w_2x_2 + b), \tag{1.8}$$

where

$$\sigma(t) = \frac{1}{1 + e^{-t}} \tag{1.9}$$

is the logistic or sigmoid function. The expression $w_1x_1 + w_2x_2 + b$ is a score. The sigmoid maps that score to a probability between 0 and 1.

For multiclass classification, a model may compute several scores, often called *logits*. For example, an image classifier might produce a cat logit, dog logit, and ship logit. Those logits can then be transformed into class probabilities. The key idea is that classification models often separate two steps: first produce class-specific scores, then convert those scores into a decision or probability distribution.

1.6 Linear and Nonlinear Models

Models can be either linear or nonlinear, and that distinction is separate from whether the task is regression or classification. A linear regression model predicts a numerical output using a linear function of the inputs,

$$\hat{y} = \beta_0 + \beta_1x_1 + \cdots + \beta_px_p.$$

With one predictor this is a line; with two predictors it is a plane; with more predictors it is a hyperplane. A linear classifier uses a similar linear score to separate or score classes, often followed by a probability transformation such as the logistic function.

Nonlinear models allow more flexible shapes. For example, a regression model might include a term such as $(x - a)^2$, and a classifier might use a curved score before converting it to a probability. The key modeling question is whether the model class can capture the structure in the data without chasing noise. Later, Chapter 11 sharpens the word “linear” by showing that transformed predictors can still lead to a model that is linear in its unknown coefficients.

1.6.1 Neural Networks as Repeated Linear and Nonlinear Operations

A fully connected neural-network layer has the form

$$\mathbf{h} = \sigma(W\mathbf{x} + \mathbf{b}). \tag{1.10}$$

Here W is a weight matrix, $\mathbf{b}$ is a bias vector, and σ is applied componentwise. Without σ , stacked layers are still only affine transformations: linear maps plus shifts. The nonlinearity is what lets neural networks represent more complicated functions.

1.7 Statistical Concepts Used Throughout the Course

This course repeatedly uses a small set of statistical ideas: expectation, variance, covariance, correlation, and bias. The full reference text gives more derivation detail; the Core Reader keeps the definitions and interpretations needed for the chapters that follow.

1.7.1 Expectation

For a discrete random variable X with possible values $x_1, \dots, x_m$ and probabilities $p_1, \dots, p_m$,

$$\mathbb{E}[X] = \sum_{i=1}^m p_i x_i. \tag{1.11}$$

Expectation is a probability-weighted average. In vector form, if $\mathbf{x}$ stores the outcomes and $\mathbf{p}$ stores the probabilities, then $\mathbb{E}[X] = \mathbf{p}^T \mathbf{x}$. This is a weighted dot product. See the Math Foundations textbook, Chapters 2, 3, and 5.

Two properties matter throughout the course:

$$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b, \quad \mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]. \quad (1.12)$$

Linearity of expectation does not require independence.

1.7.2 Variance

Variance measures average squared deviation from the mean:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2], \quad \mu_X = \mathbb{E}[X]. \quad (1.13)$$

A useful equivalent form is

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2. \quad (1.14)$$

Shifting a random variable does not change its variance, while scaling by a multiplies variance by a^2 .

1.7.3 Covariance and Correlation

Covariance measures how two random variables move together:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (1.15)$$

Correlation rescales covariance to a unitless number:

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}. \quad (1.16)$$

For observed data, correlation can be interpreted as linear alignment between centered data vectors. This connects directly to dot products, norms, and angles; see the Math Foundations textbook, Chapters 2, 3, and 5.

1.7.4 Bias

If θ is an unknown parameter and $\hat{\theta}$ is an estimator, then

$$\text{Bias}(\hat{\theta}) = \mathbb{E}[\hat{\theta}] - \theta. \quad (1.17)$$

Bias is systematic error. An estimator can be unbiased but variable, or biased but stable. Later chapters use this distinction to discuss prediction error and model flexibility.

1.8 Reproducibility and Defensibility

A data analysis must be more than numerically correct at one moment on one computer. It should also be reproducible and defensible.

1.8.1 Reproducibility

Reproducibility means that the same code applied to the same data produces the same results. A reproducible workflow has several features:

- the data source is clear;
- preprocessing steps are documented;
- transformations are recorded;
- modeling choices are traceable;
- assumptions and exclusions are stated;
- results can be rerun without guessing what happened.

A useful standard is this: your future self, or another analyst, should be able to rerun the analysis and understand the major decisions without needing a private explanation from you.

1.8.2 Defensibility

Defensibility concerns what we can explain to a customer, commander, decision maker, or technical reviewer. A defensible analysis answers questions such as:

- Why did we choose this model?
- What assumptions does the method rely on?
- What alternatives did we consider?
- What are the limitations of the data?
- How much uncertainty remains?
- What conclusion is supported, and what conclusion would overstate the evidence?

Reproducibility is about what we did. Defensibility is about what we can justify. Both are essential. A result can be reproducible but not defensible if it repeatedly applies the wrong method. A result can sound defensible but not be reproducible if the workflow cannot be checked. Good data analysis requires both.

1.9 Math Foundations Source Map

The Core Reader keeps the short Math Foundations connections where they clarify the chapter. The full Math Foundations textbook-location map for this chapter appears in the full reference textbook.

1.10 Chapter Summary

This chapter introduced data analysis as statistical learning. A data set consists of observations and variables. In supervised learning, we distinguish input variables from an output or response

variable. We model their relationship using

$$Y = f(\mathbf{X}) + \epsilon,$$

where f is the systematic relationship and ϵ is the unexplained error.

The two major modeling goals are inference and prediction. Inference asks what the model says about the relationship between inputs and output. Prediction asks how accurately the model can estimate outputs for new inputs. Regression problems involve quantitative outputs, while classification problems involve categorical outputs. Models may be linear or nonlinear, and modern methods such as neural networks build complex functions by combining linear operations with nonlinear activations.

Finally, the chapter reviewed expectation, variance, covariance, correlation, and bias. These concepts provide the mathematical language needed to discuss model error, uncertainty, assumptions, and the limits of prediction. The next chapter uses this foundation to study prediction error, model flexibility, overfitting, and generalization.

Chapter 2

Prediction Error, Flexibility, and Generalization

2.1 Why Prediction Error Matters

The previous chapter introduced the supervised learning model

$$Y = f_0(\mathbf{X}) + \epsilon, \quad (2.1)$$

where f_0 is the true systematic relationship and ϵ is unexplained error. This chapter asks a prediction-focused question:

Given the information in $\mathbf{X}$, how accurately can we predict Y ?

We write f_0 for the true unknown function and $\hat{f}$ for a fitted model. The fitted model is random because it depends on the training data. A different training sample would usually produce a different $\hat{f}$.

2.2 Mean Squared Error

For a new observation with predictors $\mathbf{x}_0$ and response Y_0 , the prediction is

$$\hat{Y}_0 = \hat{f}(\mathbf{x}_0).$$

The mean squared error at $\mathbf{x}_0$ is

$$\text{MSE}(\mathbf{x}_0) = \mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right]. \quad (2.2)$$

For observed data, the empirical MSE is

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (2.3)$$

The root mean squared error is

$$\text{RMSE} = \sqrt{\text{MSE}}, \quad (2.4)$$

which is useful because it is measured in the same units as the response.

Math Foundations Connection. Let $\mathbf{y}$ be the response vector, $\hat{\mathbf{y}}$ the prediction vector, and $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ the residual vector. Then

$$\text{MSE} = \frac{1}{n} \|\mathbf{y} - \hat{\mathbf{y}}\|^2. \quad (2.5)$$

Thus empirical MSE is a scaled squared distance. See the Math Foundations textbook, Chapters 2, 3, 5, and 7.

2.3 The Bias–Variance Decomposition

For a fixed predictor value $\mathbf{x}_0$, suppose

$$Y_0 = f_0(\mathbf{x}_0) + \epsilon_0, \quad \mathbb{E}[\epsilon_0] = 0, \quad \text{Var}(\epsilon_0) = \sigma^2. \quad (2.6)$$

Then the expected squared prediction error decomposes as

$$\mathbb{E} \left[\left(Y_0 - \hat{f}(\mathbf{x}_0) \right)^2 \right] = \text{Bias}(\hat{f}(\mathbf{x}_0))^2 + \text{Var}(\hat{f}(\mathbf{x}_0)) + \sigma^2. \quad (2.7)$$

The three terms have different meanings:

- **Bias** is systematic error from using a model class that misses the true relationship on average.
- **Model variance** is how much the fitted model changes across different training samples.
- **Noise** is irreducible error that remains even if the systematic relationship is known perfectly.

More flexible models can reduce bias, but they usually increase variance. Good prediction is a balance, not simply the most flexible fit.

2.4 Training Error, Test Error, and Flexibility

Training MSE measures performance on data used to fit the model:

$$\text{MSE}_{\text{train}} = \frac{1}{n_{\text{train}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{train}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \quad (2.8)$$

Test MSE measures performance on data not used to fit the model:

$$\text{MSE}_{\text{test}} = \frac{1}{n_{\text{test}}} \sum_{(\mathbf{x}_i, y_i) \in \mathcal{D}_{\text{test}}} \left(y_i - \hat{f}(\mathbf{x}_i) \right)^2. \quad (2.9)$$

For prediction, test error matters more because it measures generalization.

A model's *flexibility* is its ability to represent a wide range of shapes. As a useful rule of thumb,

$$\text{more flexibility} \implies \text{lower bias but higher variance.} \quad (2.10)$$

This is why training MSE usually decreases as flexibility increases, while test MSE can decrease at first and then increase.

2.5 Overfitting and Train/Test Splits

Overfitting occurs when a model follows noise or accidental patterns in the training data. The model may look good on the training sample but perform poorly on new observations. The goal is not to make training MSE as small as possible; the goal is to choose a model that generalizes.

A train/test split gives a first check on generalization:

1. split the data into training and test sets;
2. fit the model using only the training set;
3. predict the test responses;
4. compute test MSE or RMSE.

The test set must remain unseen during fitting and model tuning.

The lecture simulation used data generated from

$$Y = f_0(X) + \epsilon, \quad f_0(X) = X^3, \quad \epsilon \sim \mathcal{N}(0, 9^2). \quad (2.11)$$

A linear estimate underfit the cubic truth; a moderately flexible estimate did better; and an overly flexible estimate chased noise. Training error alone rewards flexibility, but test error rewards generalization.

2.6 Parametric and Nonparametric Estimators

A *parametric* method assumes a specific functional form with a finite set of parameters. For example,

$$f(\mathbf{X}) = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p \quad (2.12)$$

reduces learning to estimating $\beta_0, \dots, \beta_p$. This is easier to fit and interpret, but it can be biased if the assumed form is too simple. The expression above is a linear combination of input variables; see the Math Foundations textbook, Chapters 2, 3, 5, and 7.

A *nonparametric* method avoids committing to a simple fixed form for f . This can reduce bias, but it usually requires more data and can increase variance. The practical question is which approach gives defensible prediction or inference for the task.

2.7 Model Flexibility and Interpretability

Flexible models can be powerful for prediction, but they may be harder to interpret. Linear models are often less flexible, but their assumptions, coefficients, uncertainty measures, and diagnostics are easier to explain. If the goal is prediction, prioritize generalization error. If the goal is inference, prioritize interpretation, assumptions, and uncertainty quantification. If the goal includes both, state the tradeoff explicitly.

2.8 Math Foundations Source Map

The Core Reader keeps the short Math Foundations connections for MSE as squared distance and parametric models as linear combinations. The full Math Foundations textbook-location map for this chapter appears in the full reference textbook.

2.9 Chapter Summary

MSE measures average squared prediction error, and RMSE puts the score back in response units. The bias–variance decomposition separates expected prediction error into squared bias, model variance, and irreducible noise. Training error measures fit to data already seen; test error measures generalization. More flexibility can reduce bias but increase variance, so the best predictive model is usually a balance between underfitting and overfitting.

Chapter 3

Data Quality, Cleaning, and Trustworthy Variables

3.1 Why Data Quality Comes Before Modeling

Before modeling, we need to know whether the variables in the data set can be trusted. Data cleaning is not clerical work before the analysis; it is part of the analysis. A model can only learn from the variables we give it. If those variables are mislabeled, inconsistently recorded, stored in the wrong type, or contaminated by impossible values, the model may produce output that looks mathematical but is not analytically defensible.

The core question is:

What must be true about our variables before we trust them enough to summarize, visualize, or model them?

A trustworthy variable is one whose meaning, type, units, allowable values, and missing-value behavior are clear enough that another analyst could reproduce and defend the cleaning decision.

3.2 Data Tables, Observations, and Variables

A data set is often organized as a rectangular table: rows are observations and columns are variables. If all entries are numeric, we can represent the table as

$$\mathbf{X} \in \mathbb{R}^{n \times p}, \tag{3.1}$$

where n is the number of observations and p is the number of variables. The entry x_{ij} is the value of variable j for observation i .

Math Foundations Connection. A numeric variable column is a vector of observed values, and a numeric data table is a matrix whose columns are variable vectors. See the Math Foundations textbook, Chapters 2 and 7.

3.3 Variable Types

The type of a variable determines which summaries, plots, transformations, and models are meaningful.

Type	Meaning	Examples
Continuous numeric	Values vary across a range, not just isolated counts.	age, temperature, time
Discrete numeric	Values are count-like, often integers.	number of missions, defects
Ordinal	Ordered categories; distances may not be meaningful.	enlisted ranks, ratings
Categorical	Unordered labels.	service branch, platform type

A common trap is treating numeric codes as numeric variables. If service branch is coded Army = 1, Marine Corps = 2, Navy = 3, those numbers are labels, not quantities. Their average is not a meaningful “average branch.”

3.4 The Data Cleaning Workflow

The Chapter 3 lecture organized cleaning around six recurring tasks:

1. identify and handle missing values;
2. fix data types;
3. rename and standardize columns;
4. subset rows and columns;
5. detect outliers;
6. run basic sanity checks.

Each decision should be repeatable and defensible. Record what you changed, why you changed it, and what assumption the change required.

3.5 Missing Values and Imputation

Missingness is not just a software problem. It is information about how the data were collected. First ask whether missing values are encoded consistently. A zero may be a valid value in one variable and a missing-value placeholder in another: `sibling count = 0` can be real, but `weight = 0` is usually not.

Two common strategies are dropping and imputing:

Strategy	Use when	Main risk
Drop rows	Few affected rows; missingness is limited.	Losing data or introducing bias.
Drop columns	Variable is mostly missing or not needed.	Removing useful information.
Simple imputation	Mean, median, mode, or fixed value is defensible.	Distorting distributions.
Conditional imputation	Other variables help predict missing values.	Adding modeling assumptions.
Multiple imputation	Missingness requires uncertainty accounting.	More complexity.

Do not impute automatically. Sometimes “did not answer” is itself meaningful, the variable is not needed, missingness is too extensive, or imputation would create more bias than it removes.

3.6 Fixing Data Types and Standardizing Variables

A variable’s software type should match its analytic meaning. Common problems include numeric columns read as strings, categorical variables stored as integers, and dates stored as text. Fixing types is necessary because software will summarize, plot, and model variables differently depending on their type.

Column names and units should also be standardized. Names like `Miles_Driven`, `miles`, and `distance` should not represent the same idea without documentation. If units differ, fix or flag them before computing summaries.

Subsetting rows and columns is also a cleaning decision. Filtering to a population of interest, selecting variables, or creating derived variables should be recorded so the analysis can be reproduced.

3.7 Outliers and Sanity Checks

An outlier is an observation unusually far from the data or inconsistent with the overall pattern. Outliers may be errors, but they may also be real and important. Use histograms, boxplots, scatterplots, and domain knowledge before deciding whether to remove anything.

Tukey fences are a common outlier flag:

$$\text{Lower fence} = Q_1 - 1.5 \text{ IQR}, \quad \text{Upper fence} = Q_3 + 1.5 \text{ IQR}. \quad (3.2)$$

A flagged value is a reason to investigate, not an automatic deletion order.

Basic sanity checks ask:

- Are units consistent?
- Are there impossible values?
- Does the row count match expectations?
- Are duplicates present?

- Are date ranges and category labels plausible?

Prof. Alderson’s warning is the right default: be wary of taking data at face value when you do not know how it was generated.

3.8 Trustworthy Variable Checklist

Before using a variable, check:

1. What does the variable mean, and who defined it?
2. What type is it: continuous, discrete, ordinal, or categorical?
3. What are the units and valid values?
4. How are missing values encoded, and is missingness meaningful?
5. Are there impossible values, duplicates, or suspicious outliers?
6. Can the cleaning decision be reproduced and defended?

3.9 Realities of Data Cleaning

Real datasets are often large, inconsistent, and generated by processes the analyst did not control. Ideally, errors are fixed at the data-entry point. In practice, we fix what we can, document what we cannot fix, and remain alert to the possibility that a dataset is not credible enough for the intended analysis.

3.10 Math Foundations Source Map

The Core Reader keeps the short Math Foundations connections for data tables as matrices, missingness indicators, and subsetting logic. The full Math Foundations textbook-location map for this chapter appears in the full reference textbook.

3.11 Chapter Summary

Data quality comes before modeling. A trustworthy variable has a clear definition, type, unit, missing-value convention, valid range, and defensible cleaning history. Missing values, incorrect data types, inconsistent labels, outliers, and failed sanity checks can all undermine an analysis. Cleaning decisions should be repeatable, documented, and defensible.

Chapter 4

Exploratory Data Analysis: Summaries and Visual Structure

4.1 Why Summaries and Visual Structure Matter

After checking variable definitions, missingness, units, data types, impossible values, and outliers, we can begin describing what the data appear to say. Exploratory data analysis, or EDA, combines numerical summaries and plots to reveal distributional shape, relationships, subgroups, and unusual structure.

The practical question is:

How do we summarize and visualize variables in a way that is accurate, interpretable, and defensible?

4.2 A Numeric Variable as a Vector

Let $x_1, \dots, x_n$ be observed values of a numeric variable. Collect them as

$$\mathbf{x} = (x_1, \dots, x_n)^T \in \mathbb{R}^n. \quad (4.1)$$

The sample mean is

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \frac{1}{n} \mathbf{1}^T \mathbf{x}. \quad (4.2)$$

Math Foundations Connection. A variable column is a vector of observed values. Dot products, sums, centered vectors, and norms give compact ways to write means, variances, covariances, and correlations. See the Math Foundations textbook, Chapters 2, 3, and 5.

4.3 Single-Number Summary Statistics

A single-number summary answers one question about a distribution; it does not describe everything. The main families are:

Family	Question	Examples
Center	Where is a typical value?	mean, median, mode
Spread	How variable are the values?	range, IQR, variance, standard deviation
Shape	How is the distribution arranged?	skewness, kurtosis, modality
Extremes	What are the edge cases?	min, max, outlier flags

4.4 Center, Spread, Shape, and Extremes

Center. The mean is the balance point and uses every value, so it is sensitive to outliers. The median is the middle ordered value and is more robust. The mode is the most common value and is often most natural for categorical variables.

Spread. The range is $\max_i x_i - \min_i x_i$. The interquartile range is $Q_3 - Q_1$. The sample variance and standard deviation are

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2, \quad s_x = \sqrt{s_x^2}. \quad (4.3)$$

Standard deviation is easier to interpret because it has the same units as the variable.

Shape. Skewness describes asymmetry: right-skewed distributions have a long right tail, and left-skewed distributions have a long left tail. Kurtosis describes tail weight and the tendency to produce extreme values. In the Core Reader, the main point is interpretive: shape affects which summaries and models are appropriate.

Extremes. Minimums, maximums, and outlier flags identify edge cases. Tukey fences are

$$Q_1 - 1.5 \text{ IQR} \quad \text{and} \quad Q_3 + 1.5 \text{ IQR}. \quad (4.4)$$

They flag observations for investigation; they do not prove the observations are errors.

4.5 Visual EDA

Visual EDA shows patterns that one-number summaries can hide.

Data setting	Useful plots	What to look for
One categorical variable	bar plot	counts, imbalance, rare categories
One numeric variable	histogram, boxplot	skewness, modality, outliers
Categorical + numeric	grouped box plot	group differences, spread, outliers
Two numeric variables	scatterplot	trend, curvature, clusters, outliers
Many numeric variables	scatterplot matrix, heatmap	pairwise structure and correlation patterns

A histogram helps check symmetry, skewness, and modality. A scatterplot helps check whether a relationship is linear, curved, clustered, or affected by unusual points. Scatterplot matrices are useful for many pairwise relationships, but they can become visually dense. Correlation heatmaps summarize many pairwise linear relationships more compactly.

4.6 Applications of Standard Deviation

For approximately bell-shaped data, the empirical rule says that about 68%, 95%, and 99.7% of observations lie within one, two, and three standard deviations of the mean. Chebyshev's inequality is more general: for any distribution with finite variance,

$$\Pr(|X - \mu| \geq k\sigma) \leq \frac{1}{k^2}, \quad k > 1. \quad (4.5)$$

The empirical rule gives an approximate percentage under a bell-shape assumption. Chebyshev gives a conservative bound without requiring bell-shaped data.

4.7 Covariance, Correlation, and Heatmaps

Sample covariance measures how two variables vary together:

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}). \quad (4.6)$$

Sample correlation rescales covariance:

$$r_{xy} = \frac{s_{xy}}{s_x s_y}. \quad (4.7)$$

Correlation is unitless and lies between -1 and 1 when both variables have nonzero variance. For nonconstant variables, the diagonal entries of a correlation heatmap are 1 ; constant-variable correlations are undefined.

Math Foundations Connection. If $\tilde{\mathbf{x}} = \mathbf{x} - \bar{x}\mathbf{1}$ and $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}$, then correlation is a normalized dot product:

$$r_{xy} = \frac{\tilde{\mathbf{x}}^T \tilde{\mathbf{y}}}{\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|}. \quad (4.8)$$

This is why correlation measures linear alignment between centered data vectors.

4.8 A Practical EDA Workflow

A defensible EDA pass should answer:

1. What type is each variable?
2. What are the center, spread, shape, and extremes for key numeric variables?
3. Are missing values, impossible values, or suspicious zeros present?

4. Which plots reveal structure that summaries hide?
5. Which relationships look linear, curved, grouped, or outlier-driven?
6. What should be checked before modeling?

Do not treat a plot as proof. Treat it as evidence that guides the next question.

4.9 Math Foundations Source Map

The Core Reader keeps the short Math Foundations connections for vector summaries, centered vectors, norms, and correlation as a normalized dot product. The full Math Foundations textbook-location map for this chapter appears in the full reference textbook.

4.10 Chapter Summary

EDA combines summaries and visuals. Measures of center, spread, shape, and extremes answer different questions. Visual EDA reveals distributional structure, relationships, clusters, and outliers. The empirical rule and Chebyshev's inequality connect standard deviation to probability statements. Covariance and correlation summarize pairwise linear association, and heatmaps help inspect many such associations at once. Good EDA is descriptive, skeptical, and defensible.

Chapter 5

Simple Linear Regression: Model, Assumptions, and Least Squares Estimation

5.1 The Big Picture

The previous chapters built the workflow needed before formal modeling: define trustworthy variables, clean the data, summarize distributions, and visualize relationships. This chapter begins the formal regression sequence.

The central question is:

Given one input variable and one quantitative response variable, what is the equation of the best line through the data?

Simple linear regression, or SLR, is the first model we use to answer that question. It predicts a quantitative response Y using a single predictor X . The word *simple* means one predictor. The word *linear* means that the conditional mean of Y is modeled as a straight-line function of X .

SLR is important for three reasons.

1. It is the cleanest place to see the difference between a true model and an estimated model.
2. It gives explicit formulas for the least squares estimators $\hat{\beta}_0$ and $\hat{\beta}_1$.
3. It introduces residuals, normal equations, and model assumptions that appear throughout the rest of regression.

5.2 Review: From Statistical Learning to Linear Regression

In the statistical learning framework, we assume that a response variable Y is related to predictors through

$$Y = f(\mathbf{X}) + \epsilon, \tag{5.1}$$

where f is the systematic part of the relationship and ϵ is random error or noise.

If $\mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] = 0$, then

$$\mathbb{E}[Y \mid \mathbf{X} = \mathbf{x}] = \mathbb{E}[f(\mathbf{X}) + \epsilon \mid \mathbf{X} = \mathbf{x}] \quad (5.2)$$

$$= f(\mathbf{x}) + \mathbb{E}[\epsilon \mid \mathbf{X} = \mathbf{x}] \quad (5.3)$$

$$= f(\mathbf{x}). \quad (5.4)$$

Thus, under the mean-zero error condition, $f(\mathbf{x})$ is the conditional mean of Y given the input values. It is the best mean-squared-error predictor of Y given the information in $\mathbf{X}$.

In simple linear regression, there is only one predictor. We replace the general unknown function $f(X)$ with a parametric straight-line model:

$$f(X) = \beta_0 + \beta_1 X. \quad (5.5)$$

The modeling assumption becomes

$$Y = \beta_0 + \beta_1 X + \epsilon. \quad (5.6)$$

The unknown parameters are:

- β_0 , the intercept;
- β_1 , the slope.

Once we estimate these parameters from data, our fitted regression function is

$$\hat{f}(X) = \hat{\beta}_0 + \hat{\beta}_1 X. \quad (5.7)$$

In data analysis language, we say that we *regress* Y *onto* X .

5.3 Data Notation

Suppose we observe n paired data points:

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \quad (5.8)$$

Here:

- x_i is the observed predictor value for observation i ;
- y_i is the observed response value for observation i .

It is often useful to collect the observations in vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}. \quad (5.9)$$

The fitted response vector is

$$\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}, \quad \hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (5.10)$$

The residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \begin{bmatrix} e_1 \\ e_2 \\ \vdots \\ e_n \end{bmatrix}, \quad e_i = y_i - \hat{y}_i. \quad (5.11)$$

Math Foundations Connection. A variable column is a vector in $\mathbb{R}^n$; see the Math Foundations textbook, Chapters 2, 3, 5, and 12. The dot product and Euclidean norm give compact ways to write sums of products and sums of squared deviations. This is why least squares can be written both as a calculus problem and as a geometry problem.

5.4 Predicted Values, Residuals, MSE, and RSS

For any proposed line

$$\hat{y} = b_0 + b_1x, \quad (5.12)$$

where b_0 and b_1 are candidate values, the residual for observation i is

$$e_i(b_0, b_1) = y_i - (b_0 + b_1x_i). \quad (5.13)$$

The empirical mean squared error is

$$\text{MSE}(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.14)$$

The residual sum of squares is

$$\text{RSS}(b_0, b_1) = \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.15)$$

Because $\text{MSE}(b_0, b_1) = \text{RSS}(b_0, b_1)/n$, minimizing MSE and minimizing RSS give the same fitted line.

The least squares estimators are the values of b_0 and b_1 that minimize the empirical mean squared error:

$$(\hat{\beta}_0, \hat{\beta}_1) = \arg \min_{b_0, b_1} \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1x_i)]^2. \quad (5.16)$$

5.5 The Big Three Regression Assumptions

The source lectures present these as three increasingly strong assumptions. We keep that same three-step ladder throughout the regression chapters.

The model

$$Y = \beta_0 + \beta_1X + \epsilon \quad (5.17)$$

can be paired with three assumption levels. These levels matter because they determine what claims we are allowed to make.

5.5.1 Assumption 1: Mean-zero errors

The first assumption is

$$\mathbb{E}[\epsilon \mid X = x] = 0 \quad \text{for every } x. \quad (5.18)$$

Then

$$\mathbb{E}[Y \mid X = x] = \beta_0 + \beta_1 x. \quad (5.19)$$

In simple linear regression, this means that the line is the conditional mean of Y given $X = x$. In this setting, the line is a statement about average behavior, not about every individual observation.

5.5.2 Assumption 2: Constant variance and uncorrelated errors

The second assumption is stated after conditioning on the observed predictor values. In other words, we treat the predictor values as fixed and study the remaining randomness in the errors:

$$\mathbb{E}[\epsilon_i \mid X_i = x_i] = 0, \quad (5.20)$$

$$\text{Var}(\epsilon_i \mid X_i = x_i) = \sigma^2 \quad \text{for all } i, \quad (5.21)$$

$$\text{Cov}(\epsilon_i, \epsilon_j \mid X_1 = x_1, \dots, X_n = x_n) = 0 \quad \text{for } i \neq j. \quad (5.22)$$

The second condition is constant error variance, also called homoskedasticity. The third condition says that, after the predictor values are fixed, errors are uncorrelated across observations.

These assumptions support the classical standard error formulas and hypothesis tests that appear later.

5.5.3 Assumption 3: Gaussian errors

The third assumption is the Gaussian version of the same fixed-predictor idea:

$$\epsilon \mid X_1 = x_1, \dots, X_n = x_n \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (5.23)$$

Equivalently, conditional on the observed predictor values, the errors are independent normal random variables with mean zero and common variance σ^2 . This Gaussian error model supports exact finite-sample inference using t and F distributions. This assumption is powerful, but it should not be made casually. Residual diagnostics later in the course help us check whether it is plausible.

Assumption level	What it says	What it supports
Mean-zero errors	$\mathbb{E}[\epsilon \mid X = x] = 0$	Line estimates the conditional mean
Constant variance and uncorrelated errors	Conditional mean zero, constant variance, uncorrelated errors after fixing predictor values	Standard errors and approximate inference
Gaussian errors	Conditional independent normal errors with common variance	Exact classical inference under the model

Table 5.1: The three-step regression assumption ladder for simple linear regression.

5.6 Deriving the Least Squares Estimators

For the Core Reader, keep the optimization target and the key stationarity conditions. The least squares line minimizes

$$Q(b_0, b_1) = \frac{1}{n} \sum_{i=1}^n [y_i - (b_0 + b_1 x_i)]^2. \quad (5.24)$$

Setting the two partial derivatives equal to zero gives the normal equations

$$\sum_{i=1}^n [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)] = 0, \quad (5.25)$$

$$\sum_{i=1}^n x_i [y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)] = 0. \quad (5.26)$$

Equivalently,

$$\sum_{i=1}^n e_i = 0, \quad (5.27)$$

$$\sum_{i=1}^n x_i e_i = 0. \quad (5.28)$$

So least squares residuals must balance to zero overall and be orthogonal to the predictor direction.

Solving the normal equations yields

$$\boxed{\hat{\beta}_1 = \frac{\overline{xy} - \bar{x}\bar{y}}{\overline{x^2} - \bar{x}^2}} \quad (5.29)$$

and

$$\boxed{\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}.} \quad (5.30)$$

Core Reader note. The full reference build keeps the complete calculus derivation step by step. Here the main point is the logic: minimize squared residuals, use the normal equations, and solve for slope and intercept.

5.7 Equivalent Centered Form

The slope formula in (5.29) can be written in a more interpretable centered form. First,

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) = \frac{1}{n} \sum_{i=1}^n (x_i y_i - x_i \bar{y} - \bar{x} y_i + \bar{x} \bar{y}) \quad (5.31)$$

$$= \overline{xy} - \bar{x}\bar{y} - \bar{x}\bar{y} + \bar{x}\bar{y} \quad (5.32)$$

$$= \overline{xy} - \bar{x}\bar{y}. \quad (5.33)$$

Similarly,

$$\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2 = \overline{x^2} - \bar{x}^2. \quad (5.34)$$

Thus,

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}. \quad (5.35)$$

This is often the most useful formula for interpreting the slope.

5.8 Slope as Covariance Divided by Variance

The unbiased sample covariance of x and y is

$$s_{xy} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (5.36)$$

while the unbiased sample variance of x is

$$s_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2. \quad (5.37)$$

Because the same denominator $n-1$ appears in both quantities, it cancels in the ratio. Therefore,

$$\boxed{\hat{\beta}_1 = \frac{s_{xy}}{s_x^2}}. \quad (5.38)$$

Equivalently, if we use biased denominator- n versions,

$$\hat{\sigma}_{xy} = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}), \quad (5.39)$$

$$\hat{\sigma}_x^2 = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2, \quad (5.40)$$

then

$$\hat{\beta}_1 = \frac{\hat{\sigma}_{xy}}{\hat{\sigma}_x^2}. \quad (5.41)$$

For the slope, the biased-versus-unbiased denominator choice does not change the answer because the denominator cancels.

5.8.1 Interpretation of the Ratio

The numerator measures how x and y move together. The denominator measures how much x itself varies. Thus,

The least squares slope is the co-movement of x and y , scaled by the variation in x .

If x and y tend to move together, then $s_{xy} > 0$ and the slope is positive. If larger x values tend to pair with smaller y values, then $s_{xy} < 0$ and the slope is negative. If x and y have little linear co-movement, then s_{xy} is near zero and the slope is near zero.

5.9 When the Slope Is Not Identified

The formula

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (5.42)$$

requires

$$\sum_{i=1}^n (x_i - \bar{x})^2 > 0. \quad (5.43)$$

This condition means that the observed x values are not all identical.

If every observation has the same predictor value, then there is no information about how Y changes when X changes. The intercept may still describe the mean response at that one X value, but the slope cannot be estimated from the data. This is the simple-regression version of identifiability.

5.10 Matrix and Geometry View

Math Foundations Connection. The matrix notation below uses the design matrix, transposes, matrix-vector multiplication, dot products, norms, and orthogonality. These supporting concepts are covered in the Math Foundations textbook, Chapters 5 and 7.

Although this chapter derives the estimators using calculus, the same result has a clean linear algebra form. Define the design matrix

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} b_0 \\ b_1 \end{bmatrix}. \quad (5.44)$$

Then the fitted vector is

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}, \quad (5.45)$$

and the least squares objective can be written as

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2. \quad (5.46)$$

Using the dot product definition of the Euclidean norm,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}). \quad (5.47)$$

Taking derivatives in matrix form gives

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (5.48)$$

Equivalently,

$$\boxed{\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}.} \quad (5.49)$$

These are the normal equations in matrix form.

The vector equation

$$\mathbf{X}^T \mathbf{e} = \mathbf{0} \quad (5.50)$$

says that the residual vector is orthogonal to both columns of $\mathbf{X}$: the ones column and the predictor column. This is the geometric meaning of least squares. The fitted vector is the point in the line-generated model space that is closest to the observed response vector.

Bridge to Later Chapters. In the next regression geometry chapters, we will generalize this same idea: least squares projects $\mathbf{y}$ onto the subspace spanned by the columns of the design matrix.

5.11 Interpreting $\hat{\beta}_0$ and $\hat{\beta}_1$

The fitted simple linear regression model is

$$\hat{\mathbb{E}}[Y \mid X = x] = \hat{\beta}_0 + \hat{\beta}_1 x. \quad (5.51)$$

5.11.1 Intercept

The intercept $\hat{\beta}_0$ estimates the expected value of Y when $X = 0$:

$$\hat{\beta}_0 \approx \mathbb{E}[Y \mid X = 0]. \quad (5.52)$$

This interpretation is meaningful only when $X = 0$ is within the scope of the data and the model. For example, if X is years of experience, then $X = 0$ may be meaningful. If X is adult height measured in inches, then $X = 0$ is not meaningful in the observed population. In that case, the intercept is still mathematically necessary for the line, but it should not be given a literal operational interpretation.

5.11.2 Slope

The slope $\hat{\beta}_1$ estimates the change in the conditional mean of Y for a one-unit increase in X :

$$\hat{\beta}_1 \approx \mathbb{E}[Y \mid X = x + 1] - \mathbb{E}[Y \mid X = x]. \quad (5.53)$$

When the linear model is appropriate, this change is constant across x .

The units of the slope are

$$\frac{\text{units of } Y}{\text{units of } X}. \quad (5.54)$$

For example, if Y is test score and X is hours studied, then the slope is measured in test-score points per hour studied.

5.12 Association Is Not Causation

The slope is a statement about association unless the study design and assumptions justify a causal interpretation.

Suppose X is the height of mercury in a thermometer and Y is ambient temperature. A fitted regression may show a strong positive slope. But if we artificially raise the mercury level by one unit, we should not expect the ambient temperature to rise. The mercury height measures temperature; it does not cause temperature.

Before interpreting a slope causally, ask:

1. Was X randomly assigned or experimentally controlled?

2. Could another variable cause both X and Y ?
3. Is there a plausible mechanism from X to Y ?
4. Is the interpretation inside the range of observed X values?
5. Would changing X actually change Y , or only change how we measure Y ?

In this course, unless a causal design has been established, interpret regression slopes as associational.

5.13 Variance and Covariance Notation

The notation difference between variance and covariance is worth understanding.

For a single variable X , variance is a second central moment:

$$\text{Var}(X) = \mathbb{E}[(X - \mu_X)^2]. \quad (5.55)$$

It has squared units. If X is measured in meters, then $\text{Var}(X)$ is measured in square meters.

For two variables X and Y , covariance is a cross-moment:

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)]. \quad (5.56)$$

It has product units. If X is measured in meters and Y is measured in seconds, then covariance is measured in meter-seconds.

This is why the sample variance is written with a square,

$$s_x^2, \quad (5.57)$$

while sample covariance is written with two subscripts,

$$s_{xy}. \quad (5.58)$$

The square in s_x^2 signals the squared unit structure of a variance. The subscripts in s_{xy} signal co-movement between two variables.

5.14 Worked Example: Computing the Least Squares Line by Hand

Consider three observations:

x_i	y_i
1	2
2	3
3	5

The sample means are

$$\bar{x} = 2, \quad \bar{y} = \frac{10}{3}. \quad (5.59)$$

The centered slope formula gives

$$\hat{\beta}_1 = \frac{\sum_{i=1}^3 (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^3 (x_i - \bar{x})^2} = \frac{3}{2} = 1.5. \quad (5.60)$$

Then

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = \frac{1}{3}, \quad (5.61)$$

so the least squares line is

$$\hat{y} = \frac{1}{3} + 1.5x. \quad (5.62)$$

The normal-equation checks still hold:

$$\sum_{i=1}^3 e_i = 0, \quad \sum_{i=1}^3 x_i e_i = 0. \quad (5.63)$$

This is the representative hand-computation example for the Core Reader; the full reference build keeps the longer arithmetic walkthrough.

5.15 A Practical Interpretation Checklist

When interpreting a fitted SLR model,

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x, \quad (5.64)$$

work through the following checklist.

1. **Response:** What is Y , and what are its units?
2. **Predictor:** What is X , and what are its units?
3. **Intercept:** Is $X = 0$ meaningful and inside the data range?
4. **Slope:** What does “one unit of X ” mean operationally?
5. **Association:** Does the slope describe association or causation?
6. **Range:** Are predictions being made inside the observed X range?
7. **Residuals:** Are residuals small enough for the model to be useful?
8. **Assumptions:** Which assumption set is needed for the claim being made?

Coding companion reference. Package-specific Python/R commands for fitting simple linear regression and producing quick residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the derivation, interpretation checklist, and common mistakes.

5.16 Common Mistakes

1. **Dropping the parentheses in the residual.** The residual is $y_i - (b_0 + b_1x_i)$, not $y_i - b_0 + b_1x_i$.
2. **Treating the intercept as always meaningful.** The intercept is only meaningful when $X = 0$ is meaningful in context.
3. **Calling the slope causal without design support.** Regression slopes are usually associational.
4. **Ignoring units.** The slope has units of response per predictor unit.
5. **Extrapolating outside the data range.** A fitted line may behave badly outside the observed X values.
6. **Confusing residuals with true errors.** Residuals estimate errors, but they are not the unobserved errors themselves.
7. **Forgetting the denominator condition.** If all x_i values are the same, the slope cannot be estimated.

5.17 Chapter Summary

Simple linear regression estimates the conditional mean of a quantitative response using one predictor. The model is

$$Y = \beta_0 + \beta_1 X + \epsilon. \quad (5.65)$$

The least squares estimators minimize the empirical mean squared error, equivalently the residual sum of squares. Calculus gives the normal equations, and solving them yields

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{s_{xy}}{s_x^2}, \quad (5.66)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.67)$$

The slope is the sample covariance of x and y scaled by the sample variance of x . The intercept anchors the line through the sample means. The residuals sum to zero and are orthogonal to the predictor column.

The main interpretive discipline is this: least squares gives the best line under a squared-error criterion, but the usefulness of that line depends on modeling assumptions, residual behavior, variable meaning, and whether the analyst is making an associational or causal claim.

5.18 Math Foundations Source Map

This chapter's Math Foundations support is still active in the Core Reader, but the full Math Foundations textbook-location map is kept in the full reference build. The most important connections here are vectors in $\mathbb{R}^n$, dot products and norms, orthogonality, and matrix-vector multiplication.

5.19 Practice and Workflow

Can You Do This?

For Core Reader study, be sure you can do the following without looking up the full template.

1. Write the SLR model and name the assumption level needed for the claim.
2. Compute $\hat{\beta}_1$ and $\hat{\beta}_0$ from centered sums.
3. Interpret the slope with units and the intercept only when $X = 0$ is meaningful.
4. Explain why $\sum e_i = 0$ and $\sum x_i e_i = 0$ are the key algebra checks.
5. Say clearly whether the fitted relationship is associational or causal.

Chapter 6

Simple Linear Regression: Fit Quality, R^2 , and Hypothesis Testing

6.1 The Big Picture

The previous chapter built the simple linear regression model and derived the least squares line

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x. \quad (6.1)$$

The next question is not how to find the line. The next question is:

Once we have a fitted line, how good is it, and is there statistical evidence that the predictor is useful?

This chapter introduces two related but different ideas.

1. **Fit quality:** How much variation in the response does the fitted line explain? This is measured by R^2 .
2. **Statistical evidence:** Is the fitted relationship distinguishable from a zero-slope model after accounting for sample size and noise? This is tested with the F -test or, equivalently in simple linear regression, the slope t -test.

The difference matters. A model can have a high R^2 and still be based on too few observations to trust. A model can have a statistically significant slope but still explain little practical variation. Good analysis reports both fit quality and inferential evidence.

6.2 Review: The Mean-Only Baseline Model

Before using a predictor, we can always estimate a quantitative response Y by its sample mean:

$$\hat{y}_i = \bar{y}, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i. \quad (6.2)$$

This is the *mean-only model*. It ignores all predictor values.

The prediction error for observation i under the mean-only model is

$$y_i - \bar{y}. \quad (6.3)$$

The mean squared error of the mean-only model is

$$\frac{1}{n} \sum_{i=1}^n (y_i - \bar{y})^2. \quad (6.4)$$

This is the denominator- n version of sample variance. Thus, for one-dimensional data, evaluating the mean-only estimate by MSE is essentially evaluating variance.

Math Foundations Connection. A response column $(y_1, \dots, y_n)^T$ is a vector in $\mathbb{R}^n$, and the mean-only fitted vector can be written as $\bar{y}\mathbf{1}$. The sums of squared deviations below are squared Euclidean norms; see the Math Foundations textbook, Chapters 2, 3, and 5.

6.3 Adding a Predictor

When we observe paired data

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n), \quad (6.5)$$

we can compare two strategies:

1. predict all responses using $\bar{y}$;
2. predict responses using the fitted regression line

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i. \quad (6.6)$$

If X is useful, then the fitted line should have smaller squared prediction error than the mean-only model. If X is useless, then the fitted line will look like a horizontal line with slope near zero, and it will not improve much over the mean-only model.

This is the core intuition behind R^2 and the F -test:

Compare the error from the mean-only model to the error from the fitted regression model.

6.4 Three Sums of Squares

Let

$$e_i = y_i - \hat{y}_i \quad (6.7)$$

be the residual from the fitted regression line.

6.4.1 Total Sum of Squares

The *total sum of squares* measures how much the response varies around its sample mean:

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2. \quad (6.8)$$

This is the squared error from the mean-only model.

6.4.2 Residual Sum of Squares

The *residual sum of squares* measures how much variation remains after fitting the regression line:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n e_i^2. \quad (6.9)$$

This is the squared error from the fitted regression model.

6.4.3 Regression Sum of Squares

The *regression sum of squares*, sometimes called explained sum of squares, measures how much variation the model explains relative to the mean-only baseline:

$$\text{RegSS} = \text{TSS} - \text{RSS}. \quad (6.10)$$

When the regression model includes an intercept, least squares gives the decomposition

$$\boxed{\text{TSS} = \text{RegSS} + \text{RSS}.} \quad (6.11)$$

Math Foundations Connection. The decomposition in (6.11) is a geometry statement. It relies on the residual vector being orthogonal to the fitted-value direction after centering. Dot products, norms, and orthogonality are covered in the Math Foundations textbook, Chapters 2, 3, and 5.

6.5 R^2 : Proportion of Variation Explained

The coefficient of determination is

$$\boxed{R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}}.} \quad (6.12)$$

The numerator $\text{TSS} - \text{RSS}$ is the reduction in squared error achieved by using the predictor. The denominator TSS is the squared error we had before using the predictor.

Thus,

R^2 is the proportion of the response variation explained by the regression model.

If $R^2 = 0$, then the fitted model does no better than the mean-only model. If $R^2 = 1$, then the residuals are all zero and the fitted line passes exactly through every observed point.

For simple linear regression with an intercept,

$$0 \leq R^2 \leq 1. \quad (6.13)$$

This interval property depends on comparing an intercept-containing regression model to the mean-only baseline. Some nonstandard models, such as no-intercept regressions, can produce different behavior.

6.6 Interpreting R^2

Suppose two models are used to predict pediatric weight.

- Model A uses wrist circumference and has $R^2 = 0.25$.
- Model B uses height and has $R^2 = 0.50$.

Then Model B explains twice as much of the observed weight variation as Model A, at least in the data used to fit the models.

The phrase *explains variation* should be read carefully. It does not automatically imply causation. It means that the fitted line reduces squared prediction error relative to the mean-only model.

6.7 R^2 and Correlation in Simple Linear Regression

For simple linear regression with an intercept,

$$R^2 = r_{xy}^2. \quad (6.14)$$

This is a special one-predictor identity: the proportion of response variation explained by the fitted line equals the squared sample correlation between the predictor and the response. In multiple regression, there is no single ordinary predictor-response correlation that plays the same role.

6.8 Limitations of R^2

R^2 is useful, but it does not tell us everything.

6.8.1 A Perfect Fit Can Be Misleading

With two data points and a line with two parameters, we can fit the data perfectly as long as the two x values are different. Then

$$\text{RSS} = 0 \quad \Rightarrow \quad R^2 = 1. \quad (6.15)$$

This does not mean the model will predict future data well. It means only that a line can pass through two points.

6.8.2 R^2 Does Not Measure Statistical Evidence

R^2 measures in-sample fit. It does not directly answer:

- Is the slope distinguishable from zero?
- How much sampling uncertainty is in $\hat{\beta}_1$?
- How noisy are the observations?
- How many observations were used?

These questions require hypothesis tests, confidence intervals, and residual diagnostics.

6.8.3 R^2 Does Not Check Model Assumptions

A high R^2 does not prove that the linear model assumptions are valid. Residual plots may still reveal curvature, nonconstant variance, outliers, or dependence among observations.

6.9 From Fit Quality to Hypothesis Testing

Fit quality asks:

How much did the line reduce squared error compared to the mean-only model?

Hypothesis testing asks:

Is that reduction large enough, relative to residual noise and sample size, to count as statistical evidence that the slope is not zero?

For SLR, the inferential question is usually

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0. \quad (6.16)$$

The null hypothesis says that X does not improve the conditional mean of Y through a linear slope. The alternative says that the slope is nonzero.

Math Foundations Connection. Hypotheses are mathematical statements, and hypothesis testing depends on understanding logical alternatives and negation. Basic logical statements, negation, and truth-table reasoning are covered in the Math Foundations textbook, Chapter 1.

6.10 The F -Statistic

The F -statistic compares the mean-only model to the fitted regression model.

For a general nested-model comparison,

$$F = \frac{(\text{SS}_{\text{null}} - \text{SS}_{\text{full}}) / (p_{\text{full}} - p_{\text{null}})}{\text{SS}_{\text{full}} / (n - p_{\text{full}})}, \quad (6.17)$$

where:

- SS_{null} is the squared error from the simpler model;
- SS_{full} is the squared error from the larger model;
- p_{null} is the number of parameters in the simpler model;
- p_{full} is the number of parameters in the larger model.

For simple linear regression, the null model is the mean-only model, so

$$p_{\text{null}} = 1, \quad SS_{\text{null}} = \text{TSS}. \quad (6.18)$$

The full model is the fitted line, so

$$p_{\text{full}} = 2, \quad SS_{\text{full}} = \text{RSS}. \quad (6.19)$$

Therefore,

$$F = \frac{(\text{TSS} - \text{RSS})/(2 - 1)}{\text{RSS}/(n - 2)} \quad (6.20)$$

$$= \boxed{\frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n - 2)}}. \quad (6.21)$$

The numerator is the improvement over the mean-only model. The denominator is the residual variation per leftover degree of freedom.

Math Foundations Connection. The phrase *leftover degrees of freedom* is a dimension idea. The residual variation lives in directions left over after fitting the intercept and slope. Linear independence, bases, and the fact that a linearly independent set of n -vectors can have at most n elements are introduced in the Math Foundations textbook, Chapters 2, 5, and 6.

6.11 The F -Statistic in Terms of R^2

Because

$$R^2 = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}, \quad (6.22)$$

we can rewrite the SLR F -statistic as

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS}/(n - 2)} \quad (6.23)$$

$$= \frac{R^2 \text{TSS}}{(1 - R^2) \text{TSS}/(n - 2)} \quad (6.24)$$

$$= \boxed{\frac{R^2}{1 - R^2}(n - 2)}. \quad (6.25)$$

This formula explains why sample size matters. If two simple linear regressions have the same R^2 , the one with more observations will have a larger F -statistic. More data can make the same proportion of explained variation more statistically detectable.

6.12 The F -Distribution and the p-Value

Under Assumption 3 from Chapter 5—the Gaussian error model—and the null hypothesis $H_0 : \beta_1 = 0$,

$$F \sim F_{1,n-2}. \quad (6.26)$$

The first degree of freedom is the numerator degree of freedom:

$$p_{\text{full}} - p_{\text{null}} = 2 - 1 = 1. \quad (6.27)$$

The second degree of freedom is the residual degree of freedom:

$$n - p_{\text{full}} = n - 2. \quad (6.28)$$

After calculating the observed statistic F_{obs} , the p-value is

$$p\text{-value} = \Pr(F_{1,n-2} \geq F_{\text{obs}} \mid H_0). \quad (6.29)$$

A small p-value means that, if the true slope were zero and the model assumptions held, an improvement this large would be unlikely by random sampling variation alone.

6.13 Why the Assumptions Matter

The F -test relies on the same three-step regression assumption ladder introduced in Chapter 5.

1. Assumption 1 (mean-zero errors) supports the interpretation of the line as the conditional mean.
2. Assumption 2 (constant variance and uncorrelated errors) supports the standard error and variance calculations.
3. Assumption 3 (Gaussian errors) supports exact finite-sample t and F distributions.

If these assumptions are badly violated, the p-value may not have the advertised interpretation. This is why residual analysis is not optional.

6.14 The Slope t -Test

Most software output also reports a t -test for the slope. The slope test is

$$H_0 : \beta_1 = 0 \quad \text{versus} \quad H_A : \beta_1 \neq 0. \quad (6.30)$$

The residual variance estimator is

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - 2}. \quad (6.31)$$

The standard error of the slope is

$$\text{SE}(\hat{\beta}_1) = \sqrt{\frac{\hat{\sigma}^2}{\sum_{i=1}^n (x_i - \bar{x})^2}} = \sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}. \quad (6.32)$$

The t -statistic is

$$t_{\text{obs}} = \frac{\hat{\beta}_1}{\text{SE}(\hat{\beta}_1)}. \quad (6.33)$$

Under the null hypothesis and Assumption 3 (Gaussian errors),

$$t_{\text{obs}} \sim t_{n-2}. \quad (6.34)$$

6.15 Why the SLR F -Test and Slope t -Test Agree

In simple linear regression with an intercept,

$$F = t^2. \quad (6.35)$$

Both tests answer the same question: is the slope different from zero? The F -test compares the fitted line to the mean-only model, while the slope t -test standardizes $\hat{\beta}_1$ by its standard error. In MLR these are no longer the same question: the overall F -test asks whether at least one slope is useful, while individual t -tests ask about one coefficient at a time.

6.16 Worked Example

For the five observations in the full reference example, the fitted line is

$$\hat{y} = 1.3 + 0.9x. \quad (6.36)$$

The total sum of squares is $\text{TSS} = 10$, the residual sum of squares is $\text{RSS} = 1.9$, and therefore

$$R^2 = 1 - \frac{1.9}{10} = 0.81. \quad (6.37)$$

The model explains 81% of the observed variation in y . The corresponding F -statistic is about 12.79, compared to an $F_{1,3}$ distribution under $H_0 : \beta_1 = 0$. The full reference text shows the residual table and arithmetic details.

6.17 How to Read Software Output

Regression software usually reports both fit-quality and hypothesis-testing information.

6.17.1 Fit Quality Fields

Common fields include:

- **R-squared:** the R^2 value;
- **Residual standard error** or **scale:** an estimate of residual noise;
- **F-statistic:** the model-versus-null comparison;
- **Prob(F-statistic):** the p-value for the F -test.

6.17.2 Coefficient Table Fields

For the predictor row, common fields include:

- **coef:** $\hat{\beta}_1$;
- **std err:** $\text{SE}(\hat{\beta}_1)$;
- **t:** the slope t -statistic;
- **P<—t—:** the two-sided p-value for $H_0 : \beta_1 = 0$;
- **confidence interval:** an interval estimate for the slope.

In SLR, the p-value in the predictor row and the p-value for the model F -statistic correspond to the same slope question.

6.18 Decision Template for SLR Fit and Significance

When reporting a simple linear regression, keep the distinction between fit quality and statistical evidence clear:

1. state the fitted model $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$;
2. interpret the slope with units;
3. report R^2 as an in-sample fit measure;
4. state the slope hypothesis test $H_0 : \beta_1 = 0$ versus $H_A : \beta_1 \neq 0$;
5. report the test statistic and p-value;
6. check residuals before making final claims.

Coding companion reference. Package-specific Python/R commands for computing R^2 , F -statistics, p-values, and related output have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the conceptual guidance for reading software output.

6.19 Common Mistakes

1. **Treating high R^2 as proof of causation.** R^2 is about explained variation, not causal mechanism.
2. **Treating high R^2 as proof of validity.** A model can have high R^2 and still violate assumptions.
3. **Ignoring sample size.** The same R^2 can produce different statistical evidence depending on n .
4. **Confusing residual variance with response variance.** TSS measures response variation before the model; RSS measures variation left after the model.

5. **Forgetting degrees of freedom.** In SLR, residual variance uses denominator $n - 2$ because two parameters are estimated.
6. **Thinking $R^2 = 1$ always means a useful model.** Two points can give a perfect line, but not a reliable model.
7. **Reading p-values without context.** A small p-value does not tell you effect size, practical importance, or model adequacy.

6.20 Chapter Summary

The mean-only model predicts every response by $\bar{y}$. Its squared error is TSS. The regression model predicts each response by $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$. Its remaining squared error is RSS. The key fit measure is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}, \quad (6.38)$$

which is the proportion of observed response variation explained by the fitted line.

The F -test compares the fitted line to the mean-only model. Under $H_0 : \beta_1 = 0$ and the Gaussian-error rung of the regression assumption ladder,

$$F = \frac{\text{TSS} - \text{RSS}}{\text{RSS} / (n - 2)} \sim F_{1, n-2}. \quad (6.39)$$

In SLR with an intercept, the slope t -test and model F -test agree because $F = t^2$.

Can you do this? You should be able to compute TSS, RSS, R^2 , and the SLR F -statistic; explain the difference between fit and statistical evidence; and read the slope row and model F -test from software output. The full reference text contains the complete Math Foundations source map and reusable study template.

Chapter 7

Linear Regression as Projection: Design Matrix and Hat Matrix

7.1 The Big Picture

The last two chapters developed simple linear regression using calculus, sums of squares, and hypothesis tests. This chapter rebuilds the same least squares estimator using linear algebra. The payoff is substantial:

Least squares regression is an orthogonal projection of the response vector onto the subspace spanned by the columns of the design matrix.

This viewpoint explains why the normal equations appear, why residuals are orthogonal to fitted values, why the fitted values can be written with the hat matrix, and why the same formula works for multiple linear regression.

The main objects in this chapter are:

$$\mathbf{y} \in \mathbb{R}^n \quad \text{the response vector,} \quad (7.1)$$

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)} \quad \text{the design matrix,} \quad (7.2)$$

$$\hat{\boldsymbol{\beta}} \in \mathbb{R}^{p+1} \quad \text{the least squares coefficient vector,} \quad (7.3)$$

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} \quad \text{the fitted-value vector,} \quad (7.4)$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad \text{the residual vector,} \quad (7.5)$$

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \quad \text{the hat matrix.} \quad (7.6)$$

The key formula is

$$\boxed{\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}.} \quad (7.7)$$

The matrix $\mathbf{H}$ is called the hat matrix because it puts the hat on $\mathbf{y}$: it turns observed responses into fitted responses.

Notation bridge. From this chapter onward, bold lowercase vectors such as $\mathbf{y}$ denote observed sample vectors from the data in hand, while bold uppercase symbols such as $\mathbf{X}$ denote matrices.

Chapter 10 temporarily writes $\mathbf{Y}$ when the full response vector is being treated as random conditional on a fixed design matrix. The front-matter notation guide gives the book-wide convention.

Math Foundations Connection. This chapter uses vector notation in $\mathbb{R}^n$, the ones vector $\mathbf{1}$, linear combinations, dot products, norms, orthogonality, matrix notation, transpose notation, and matrix-vector multiplication. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, and 7.

7.2 Vectors Behind a Regression Scatterplot

A simple linear regression scatterplot displays points

$$(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n). \quad (7.8)$$

In matrix language, we separate the observed predictor values and response values into vectors:

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}, \quad \mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}. \quad (7.9)$$

The scatterplot is drawn in the usual (x, y) -plane, but the regression calculation takes place in $\mathbb{R}^n$. The response vector $\mathbf{y}$ is a point or vector in n -dimensional observation space. The fitted vector $\hat{\mathbf{y}}$ is another vector in that same space.

For simple linear regression, every fitted vector has the form

$$\hat{\mathbf{y}} = \hat{\beta}_0 \mathbf{1} + \hat{\beta}_1 \mathbf{x}. \quad (7.10)$$

Thus, the fitted values live in the set of all linear combinations of $\mathbf{1}$ and $\mathbf{x}$. This set is the plane

$$\text{Col}(\mathbf{X}) = \text{span}\{\mathbf{1}, \mathbf{x}\} \quad (7.11)$$

inside $\mathbb{R}^n$, provided $\mathbf{1}$ and $\mathbf{x}$ are linearly independent.

Important interpretation. A regression line in the scatterplot corresponds to a vector $\hat{\mathbf{y}}$ on a plane in $\mathbb{R}^n$. We are not projecting points in the two-dimensional scatterplot. We are projecting the full response vector $\mathbf{y}$ onto the subspace of fitted-value vectors.

7.3 The Linear Regression Problem as an Overdetermined System

Suppose we have three data points. A perfect simple linear model would satisfy

$$y_1 = \beta_0 + \beta_1 x_1, \quad (7.12)$$

$$y_2 = \beta_0 + \beta_1 x_2, \quad (7.13)$$

$$y_3 = \beta_0 + \beta_1 x_3. \quad (7.14)$$

There are three equations and only two unknowns. With more than two points, the system is usually overdetermined. In vector form, the exact system would be

$$\mathbf{y} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}. \quad (7.15)$$

Usually $\mathbf{y} \notin \text{span}\{\mathbf{1}, \mathbf{x}\}$. Therefore, no exact solution exists.

Least squares solves the nearest possible problem:

Find the vector in $\text{span}\{\mathbf{1}, \mathbf{x}\}$ that is closest to $\mathbf{y}$ in squared Euclidean distance.

That nearest vector is the orthogonal projection of $\mathbf{y}$ onto the fitted-value plane.

7.4 Projection Onto One Vector

Before projecting onto the two-dimensional regression plane, recall projection onto a single nonzero vector $\mathbf{a}$:

$$\text{proj}_{\mathbf{a}}(\mathbf{y}) = \frac{\mathbf{a}^T \mathbf{y}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}. \quad (7.16)$$

The key fact is geometric: the leftover vector

$$\mathbf{r} = \mathbf{y} - \text{proj}_{\mathbf{a}}(\mathbf{y}) \quad (7.17)$$

is orthogonal to $\mathbf{a}$. Least squares regression keeps the same logic, but the target is now the entire column space of the design matrix rather than a single line.

7.5 The Design Matrix for Simple Linear Regression

For simple linear regression, define the design matrix

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}] = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix} \in \mathbb{R}^{n \times 2}. \quad (7.18)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \quad (7.19)$$

Then the simple linear regression model can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (7.20)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.21)$$

The columns of $\mathbf{X}$ are directions in $\mathbb{R}^n$. In SLR, those directions are $\mathbf{1}$ and $\mathbf{x}$. The fitted vector $\hat{\mathbf{y}}$ is a linear combination of those columns.

7.6 The Normal Equations from Orthogonality

The least squares residual vector is

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.22)$$

The orthogonal projection condition says that the residual vector must be orthogonal to every column of $\mathbf{X}$. For SLR, this means

$$\mathbf{1}^T \mathbf{e} = 0, \quad (7.23)$$

$$\mathbf{x}^T \mathbf{e} = 0. \quad (7.24)$$

Stacking these two equations gives

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (7.25)$$

Substitute $\mathbf{e} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}, \quad (7.26)$$

$$\mathbf{X}^T \mathbf{y} - \mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{0}, \quad (7.27)$$

$$\boxed{\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}.} \quad (7.28)$$

These are the normal equations.

If $\mathbf{X}^T \mathbf{X}$ is invertible, then

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.} \quad (7.29)$$

The invertibility condition is equivalent to the columns of $\mathbf{X}$ being linearly independent. For SLR with an intercept, this requires the observed x_i 's not all being equal.

Math Foundations Connection. The step from $\mathbf{X}^T \mathbf{e} = \mathbf{0}$ to the normal equations uses matrix transpose notation and matrix-vector multiplication. Matrix-vector multiplication can be read both as row inner products and as column linear combinations; see the Math Foundations textbook, Chapters 3, 5, 7, and 12.

Core Reader note. Chapter 5 already derived the familiar scalar formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$. The Core Reader keeps the matrix viewpoint here and leaves the full algebraic unpacking to the reference build.

7.7 The Hat Matrix

Start with the fitted values:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.30)$$

Now substitute the least squares estimator from (7.29):

$$\hat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (7.31)$$

Define

$$\boxed{\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T.} \quad (7.32)$$

Then

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}. \quad (7.33)$$

The hat matrix is an $n \times n$ matrix. It maps observed response vectors in $\mathbb{R}^n$ to fitted-value vectors in the column space of $\mathbf{X}$:

$$\mathbf{H} : \mathbb{R}^n \rightarrow \text{Col}(\mathbf{X}). \quad (7.34)$$

7.8 Properties of the Hat Matrix

For the Core Reader, keep the three high-yield facts about $\mathbf{H}$:

1. **Symmetry:** $\mathbf{H}^T = \mathbf{H}$.
2. **Idempotence:** $\mathbf{H}^2 = \mathbf{H}$.
3. **Projection meaning:** $\mathbf{H}$ is the orthogonal projection onto $\text{Col}(\mathbf{X})$.

So

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y} \quad (7.35)$$

is the closest vector to $\mathbf{y}$ inside the fitted-value space. Once a vector has already been projected there, applying $\mathbf{H}$ again does nothing.

Because

$$\hat{y}_i = \sum_{j=1}^n H_{ij}y_j, \quad (7.36)$$

the entries of $\mathbf{H}$ tell us how observed responses influence fitted responses. The diagonal entries H_{ii} are the leverage values.

Core Reader note. The full reference build keeps the step-by-step symmetry and idempotence proofs.

7.9 The Residual Maker Matrix

The hat matrix makes fitted values. The complementary matrix makes residuals.

Define

$$\mathbf{M} = \mathbf{I} - \mathbf{H}. \quad (7.37)$$

Then

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad (7.38)$$

$$= \mathbf{y} - \mathbf{H}\mathbf{y} \quad (7.39)$$

$$= (\mathbf{I} - \mathbf{H})\mathbf{y} \quad (7.40)$$

$$= \mathbf{M}\mathbf{y}. \quad (7.41)$$

The matrix $\mathbf{M}$ is also symmetric and idempotent:

$$\mathbf{M}^T = \mathbf{M}, \quad \mathbf{M}^2 = \mathbf{M}. \quad (7.42)$$

It projects $\mathbf{y}$ onto the residual subspace, which is the orthogonal complement of $\text{Col}(\mathbf{X})$.

Because $\mathbf{X}^T \mathbf{e} = \mathbf{0}$, the residual vector is orthogonal to every column of the design matrix. Since $\hat{\mathbf{y}} \in \text{Col}(\mathbf{X})$, the residual vector is also orthogonal to the fitted-value vector:

$$\hat{\mathbf{y}}^T \mathbf{e} = 0. \quad (7.43)$$

This gives the geometric decomposition

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}, \quad \hat{\mathbf{y}} \perp \mathbf{e}. \quad (7.44)$$

7.10 Dimensions and Degrees of Freedom

If $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$ has full column rank, then

$$\dim \text{Col}(\mathbf{X}) = p + 1. \quad (7.45)$$

The fitted values live in this $(p + 1)$ -dimensional subspace. The residuals live in the orthogonal complement, whose dimension is

$$n - (p + 1) = n - p - 1. \quad (7.46)$$

This is the linear algebra reason that the residual variance estimator in multiple linear regression divides by $n - p - 1$.

For SLR, $p = 1$, so the fitted-value subspace has dimension 2, and the residual subspace has dimension

$$n - 2. \quad (7.47)$$

Math Foundations Connection. The dimension claim depends on linear independence and bases. A set of linearly independent n -vectors can have at most n elements, and a basis provides unique linear-combination coordinates. See the Math Foundations textbook, Chapters 2 and 6.

7.11 Multiple Linear Regression as the Same Projection

The design matrix generalizes immediately from simple linear regression to multiple linear regression. With predictors $\mathbf{x}_1, \dots, \mathbf{x}_p$, define

$$\mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_p] = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}. \quad (7.48)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}. \quad (7.49)$$

The model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (7.50)$$

The least squares solution is still

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad (7.51)$$

and the fitted values are still

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (7.52)$$

The only change is the subspace. In SLR, $\text{Col}(\mathbf{X})$ is usually a two-dimensional plane in $\mathbb{R}^n$. In MLR, $\text{Col}(\mathbf{X})$ is usually a $(p+1)$ -dimensional subspace in $\mathbb{R}^n$.

7.12 Design Matrix as a Modeling Choice

The design matrix tells the model what directions it is allowed to use to approximate $\mathbf{y}$. Choosing a model means choosing columns for $\mathbf{X}$.

Examples:

$$\text{SLR:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}], \quad (7.53)$$

$$\text{Polynomial model:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x} \quad \mathbf{x}^2 \quad \mathbf{x}^3], \quad (7.54)$$

$$\text{MLR:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \cdots \quad \mathbf{x}_p], \quad (7.55)$$

$$\text{Interaction model:} \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}_1 \quad \mathbf{x}_2 \quad \mathbf{x}_1 \odot \mathbf{x}_2]. \quad (7.56)$$

Here $\odot$ means elementwise multiplication, so the interaction column has entries $x_{i1}x_{i2}$.

The same least squares machinery works for all of these models as long as the model is linear in the coefficients. This is why regression can fit curved relationships using transformed columns while still being a linear model in parameters.

Coding companion reference. Package-specific Python/R commands for building design matrices, hat matrices, fitted values, residuals, and leverage values have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the projection geometry and matrix interpretation.

7.13 Common Mistakes

1. **Projecting the scatterplot instead of the response vector.** The projection takes place in $\mathbb{R}^n$, not in the two-dimensional scatterplot.
2. **Forgetting the intercept column.** With an intercept, the first column of $\mathbf{X}$ is $\mathbf{1}$. Removing the intercept changes the projection subspace.
3. **Assuming $(\mathbf{X}^T \mathbf{X})^{-1}$ always exists.** It exists only when the columns of $\mathbf{X}$ are linearly independent.
4. **Confusing random errors and residuals.** The model errors $\boldsymbol{\epsilon}$ are unobserved. The residuals $\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}$ are computed after fitting the model.

5. **Thinking the hat matrix estimates coefficients.** The hat matrix maps $\mathbf{y}$ to $\hat{\mathbf{y}}$. The coefficient estimator is $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$.
6. **Ignoring design-matrix columns.** Transformed predictors, interaction terms, and categorical encodings are all columns of $\mathbf{X}$.
7. **Treating high leverage as a residual problem only.** Leverage comes from $\mathbf{X}$, not from y . It measures unusual predictor geometry.

7.14 Chapter Summary

Simple linear regression can be written as

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbf{X} = [\mathbf{1} \quad \mathbf{x}]. \quad (7.57)$$

The fitted values are linear combinations of the columns of $\mathbf{X}$:

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (7.58)$$

Least squares chooses the fitted vector in $\text{Col}(\mathbf{X})$ closest to $\mathbf{y}$. The residual vector is orthogonal to every column of $\mathbf{X}$, so

$$\mathbf{X}^T(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (7.59)$$

This gives the normal equations

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}. \quad (7.60)$$

If $\mathbf{X}^T \mathbf{X}$ is invertible, then

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}. \quad (7.61)$$

The fitted values can be written as

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (7.62)$$

The hat matrix is symmetric and idempotent, so it is an orthogonal projection matrix. The diagonal entries of $\mathbf{H}$ are leverage values, and the full matrix entries describe how observed responses influence fitted responses.

The same projection formula generalizes immediately to multiple linear regression by adding more columns to the design matrix. In MLR, when $\mathbf{X}$ has full column rank, the fitted values are the projection of $\mathbf{y}$ onto the $(p + 1)$ -dimensional column space of $\mathbf{X}$, and the residuals live in the $(n - p - 1)$ -dimensional orthogonal complement. If $\text{rank}(\mathbf{X}) = r$, replace $p + 1$ by r and $n - p - 1$ by $n - r$. Chapter 8 uses these dimensions to explain degrees of freedom and identifiability.

7.15 Math Foundations Source Map

The Core Reader keeps the geometry ideas from Math Foundations but omits the full Math Foundations textbook-location map. The main supporting ideas here are vectors in $\mathbb{R}^n$, dot products, orthogonality, bases, full rank, and least-squares normal equations.

7.16 Practice and Workflow

Can You Do This?

For Core Reader study, be sure you can do the following.

1. Build $\mathbf{X}$ with the intercept column in the correct place.
2. Explain why least squares is a projection of $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$.
3. Write the normal equations and the coefficient formula.
4. Write $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ and $\mathbf{e} = (\mathbf{I} - \mathbf{H})\mathbf{y}$.
5. State the residual degrees of freedom $n - p - 1$.

Chapter 8

Degrees of Freedom, Subspaces, and Identifiability

8.1 The Big Picture

The previous chapter showed that linear regression is an orthogonal projection. The response vector $\mathbf{y} \in \mathbb{R}^n$ is projected onto the column space of the design matrix $\mathbf{X}$. The fitted values live in one subspace, and the residuals live in the orthogonal complement.

This chapter uses that geometry to explain three ideas that appear repeatedly in regression:

1. **Degrees of freedom:** how many independent dimensions remain after we estimate model parameters.
2. **Subspaces:** where fitted values, residuals, coefficient information, and non-identifiable directions live.
3. **Identifiability:** when regression coefficients are uniquely determined by the data, and when many coefficient vectors produce the same fitted values.

The central message is:

Degrees of freedom are leftover dimensions. Identifiability fails when the design matrix has a nonzero null space.

For a full-rank multiple linear regression model with p predictors and an intercept,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \text{rank}(\mathbf{X}) = p + 1, \quad (8.1)$$

and the residual degrees of freedom are

$$\boxed{n - p - 1}. \quad (8.2)$$

This is why the denominator of the residual variance estimator and the denominator degrees of freedom in the overall regression F -test use $n - p - 1$.

Math Foundations Connection. This chapter uses vector notation, the ones vector, dot products, orthogonality, matrix dimensions, matrix transposes, linear independence, bases, matrix-vector multiplication, left-inverse intuition, and null-space/subspace ideas. These are covered in the Math Foundations textbook, Chapters 2, 5, 6, 7, 10, and 11.

8.2 A First Definition: Degrees of Freedom as Independent Movement

In data analysis, degrees of freedom are often introduced as a formula to memorize. For example:

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i - \bar{y})^2. \quad (8.3)$$

The natural question is: why $n-1$?

The geometric answer is that n observations define a vector in $\mathbb{R}^n$. Before estimating anything, the vector $\mathbf{y}$ has n independent coordinates:

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \in \mathbb{R}^n. \quad (8.4)$$

Once we estimate the mean $\bar{y}$, the residual deviations must satisfy one linear constraint:

$$\sum_{i=1}^n (y_i - \bar{y}) = 0. \quad (8.5)$$

One independent constraint removes one dimension. Therefore, the residual variation around the sample mean has $n-1$ degrees of freedom.

Reusable intuition. Each independently estimated parameter usually consumes one dimension. The leftover dimensions are available for residual variation.

8.3 Decomposing a Data Vector Around the Sample Mean

The simplest projection problem is the sample mean. Let

$$\mathbf{1} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \in \mathbb{R}^n. \quad (8.6)$$

The sample mean vector is

$$\bar{y}\mathbf{1}. \quad (8.7)$$

The residual vector around the sample mean is

$$\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}. \quad (8.8)$$

Therefore,

$$\boxed{\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{y} - \bar{y}\mathbf{1})}. \quad (8.9)$$

This is a decomposition of the data vector into two orthogonal pieces:

1. $\bar{y}\mathbf{1}$, the projection onto the one-dimensional subspace $\text{span}\{\mathbf{1}\}$, and
2. $\mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1}$, the residual vector in the orthogonal complement.

To verify orthogonality, compute

$$\mathbf{1}^T \mathbf{r} = \mathbf{1}^T (\mathbf{y} - \bar{y}\mathbf{1}) \quad (8.10)$$

$$= \sum_{i=1}^n y_i - \bar{y} \sum_{i=1}^n 1 \quad (8.11)$$

$$= n\bar{y} - n\bar{y} \quad (8.12)$$

$$= 0. \quad (8.13)$$

Thus,

$$\mathbf{r} \in \text{span}\{\mathbf{1}\}^\perp. \quad (8.14)$$

Since $\text{span}\{\mathbf{1}\}$ has dimension 1, its orthogonal complement in $\mathbb{R}^n$ has dimension

$$n - 1. \quad (8.15)$$

This is the geometric reason for the $n - 1$ denominator in sample variance.

8.4 Worked Example: Mean Decomposition with Three Observations

For $\mathbf{y} = (2, 5, 8)^T$, the sample mean is $\bar{y} = 5$. Therefore

$$\bar{y}\mathbf{1} = \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix}, \quad \mathbf{r} = \mathbf{y} - \bar{y}\mathbf{1} = \begin{bmatrix} -3 \\ 0 \\ 3 \end{bmatrix}. \quad (8.16)$$

The residuals sum to zero, so they satisfy one linear constraint. With three observations, that leaves $3 - 1 = 2$ residual degrees of freedom.

8.5 The Mean Projection Matrix

The mean-only fit can also be written as a projection:

$$\hat{\mathbf{y}}_0 = \bar{y}\mathbf{1} = \mathbf{H}_0 \mathbf{y}, \quad \mathbf{H}_0 = \frac{1}{n} \mathbf{1}\mathbf{1}^T. \quad (8.17)$$

This is the one-dimensional version of the hat-matrix idea. The full reference text develops the symmetry, idempotence, rank, and trace details.

8.6 Regression Decomposes $\mathbb{R}^n$ Into Two Orthogonal Pieces

For multiple linear regression with an intercept and p predictors, write

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \mathbf{X} \in \mathbb{R}^{n \times (p+1)}. \quad (8.18)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T, \quad (8.19)$$

when $\mathbf{X}$ has full column rank.

The residuals are

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{M}\mathbf{y}. \quad (8.20)$$

Regression decomposes the observation space as

$$\boxed{\mathbb{R}^n = \text{Col}(\mathbf{X}) \oplus \text{Null}(\mathbf{X}^T).} \quad (8.21)$$

Here $\oplus$ means a direct sum: every vector in $\mathbb{R}^n$ can be written uniquely as a piece in $\text{Col}(\mathbf{X})$ plus a piece in $\text{Null}(\mathbf{X}^T)$. These two pieces are orthogonal.

In regression terms:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}), \quad (8.22)$$

$$\mathbf{e} \in \text{Null}(\mathbf{X}^T), \quad (8.23)$$

$$\hat{\mathbf{y}}^T \mathbf{e} = 0. \quad (8.24)$$

The fitted values live in the column space. The residuals live in the left null space.

8.7 Residual Degrees of Freedom

Assume $\mathbf{X}$ has full column rank. Then

$$\text{rank}(\mathbf{X}) = p + 1. \quad (8.25)$$

Therefore,

$$\dim \text{Col}(\mathbf{X}) = p + 1. \quad (8.26)$$

The column space consumes $p + 1$ dimensions of the original n -dimensional observation space. The leftover dimensions are the residual degrees of freedom:

$$\boxed{\text{df}_{\text{resid}} = n - (p + 1) = n - p - 1.} \quad (8.27)$$

This explains the residual variance estimator:

$$\hat{\sigma}^2 = \frac{\text{RSS}}{n - p - 1} = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1}. \quad (8.28)$$

It also explains the denominator degrees of freedom in the overall F -test for MLR.

Special cases.

Model	Fitted subspace dimension	Residual df
Mean-only model	1	$n - 1$
SLR with intercept	2	$n - 2$
MLR with intercept and p predictors	$p + 1$	$n - p - 1$
Full rank model with design rank r	r	$n - r$

8.8 A Subtle Point: Model Degrees of Freedom Versus Fitted-Subspace Dimension

Two counts appear in regression. The fitted subspace has dimension $p + 1$ when the model has an intercept and p predictors. But the overall regression F -test compares the full model to the intercept-only model, so its numerator degrees of freedom are p , not $p + 1$. The residual degrees of freedom are $n - p - 1$.

8.9 Trace of the Hat Matrix

For a full-rank design matrix, the hat matrix projects onto $\text{Col}(\mathbf{X})$. Its trace equals the fitted-subspace dimension:

$$\text{tr}(\mathbf{H}) = \text{rank}(\mathbf{H}) = p + 1. \quad (8.29)$$

Thus the average leverage is $(p + 1)/n$, and the residual-maker matrix has trace $n - p - 1$, matching the residual degrees of freedom.

8.10 The Big Four Subspaces of the Design Matrix

Let

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)} \quad (8.30)$$

with rank r . Linear algebra associates four fundamental subspaces with $\mathbf{X}$. In regression, these subspaces have direct meanings.

Subspace	Where it lives	Regression meaning	Dimension
$\text{Col}(\mathbf{X})$	$\mathbb{R}^n$	Fitted values $\hat{\mathbf{y}}$	r
$\text{Null}(\mathbf{X}^T)$	$\mathbb{R}^n$	Residual vectors $\mathbf{e}$	$n - r$
$\text{Row}(\mathbf{X}) = \text{Col}(\mathbf{X}^T)$	$\mathbb{R}^{p+1}$	Identifiable coefficient directions	r
$\text{Null}(\mathbf{X})$	$\mathbb{R}^{p+1}$	Non-identifiable coefficient directions	$(p + 1) - r$

The first two subspaces live in observation space $\mathbb{R}^n$. The last two live in coefficient space $\mathbb{R}^{p+1}$. This distinction prevents a common mistake: residuals and fitted values are vectors with n entries, while coefficient directions have $p + 1$ entries.

8.11 Column Space and Left Null Space

The fitted values satisfy

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (8.31)$$

Therefore, $\hat{\mathbf{y}}$ is a linear combination of the columns of $\mathbf{X}$:

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}). \quad (8.32)$$

The residual vector satisfies the normal equations:

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (8.33)$$

Therefore,

$$\mathbf{e} \in \text{Null}(\mathbf{X}^T). \quad (8.34)$$

Since $\text{Null}(\mathbf{X}^T)$ is the orthogonal complement of $\text{Col}(\mathbf{X})$, every residual vector is orthogonal to every fitted-value direction.

This is the geometric core of regression diagnostics:

$$\mathbf{y} = \hat{\mathbf{y}} + \mathbf{e}, \quad \hat{\mathbf{y}} \perp \mathbf{e}. \quad (8.35)$$

8.12 Row Space and Null Space

The row space and null space of $\mathbf{X}$ live in coefficient space. They answer a different question:

Which coefficient directions change fitted values, and which coefficient directions are invisible to the data?

A vector $\mathbf{v} \in \mathbb{R}^{p+1}$ is in the null space of $\mathbf{X}$ if

$$\mathbf{X}\mathbf{v} = \mathbf{0}. \quad (8.36)$$

If $\mathbf{v} \neq \mathbf{0}$, then changing the coefficient vector in the direction $\mathbf{v}$ does not change fitted values:

$$\mathbf{X}(\boldsymbol{\beta} + c\mathbf{v}) = \mathbf{X}\boldsymbol{\beta} + c\mathbf{X}\mathbf{v} = \mathbf{X}\boldsymbol{\beta} \quad (8.37)$$

for any scalar c . Therefore, $\mathbf{v}$ is a non-identifiable direction.

The row space contains coefficient directions that the data can see. The null space contains coefficient directions that the data cannot see.

8.13 Identifiability

A parameter is identifiable when different parameter values imply different observable behavior. In linear regression, the coefficient vector β is identifiable when the mapping

$$\beta \mapsto \mathbf{X}\beta \quad (8.38)$$

is one-to-one.

This happens exactly when

$$\text{Null}(\mathbf{X}) = \{\mathbf{0}\}. \quad (8.39)$$

Equivalently, the columns of $\mathbf{X}$ must be linearly independent.

For $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, the following statements are equivalent:

1. β is identifiable.
2. $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$.
3. The columns of $\mathbf{X}$ are linearly independent.
4. $\text{rank}(\mathbf{X}) = p + 1$.
5. $\mathbf{X}^T \mathbf{X}$ is invertible.
6. The least squares coefficient vector

$$\hat{\beta} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} \quad (8.40)$$

is unique.

Why $\mathbf{X}^T \mathbf{X}$ matters. For any $\mathbf{v}$,

$$\mathbf{v}^T \mathbf{X}^T \mathbf{X} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2. \quad (8.41)$$

If $\mathbf{X} \mathbf{v} = \mathbf{0}$ for some nonzero $\mathbf{v}$, then $\mathbf{X}^T \mathbf{X}$ cannot be invertible. Conversely, if $\mathbf{X}^T \mathbf{X}$ is singular, then there is a nonzero $\mathbf{v}$ such that $\mathbf{X} \mathbf{v} = \mathbf{0}$. Thus, $\mathbf{X}^T \mathbf{X}$ is invertible exactly when $\mathbf{X}$ has no nonzero null directions.

8.14 What Fails When the Model Is Not Identified?

When $\text{Null}(\mathbf{X}) \neq \{\mathbf{0}\}$, there are infinitely many coefficient vectors that produce the same fitted values. If $\mathbf{v} \in \text{Null}(\mathbf{X})$, then

$$\mathbf{X} \hat{\beta} = \mathbf{X}(\hat{\beta} + c\mathbf{v}) \quad (8.42)$$

for every scalar c . The data cannot distinguish $\hat{\beta}$ from $\hat{\beta} + c\mathbf{v}$.

Two important consequences follow:

1. The coefficient vector is not unique.
2. The fitted values may still be unique, because the projection of $\mathbf{y}$ onto $\text{Col}(\mathbf{X})$ is unique.

Thus, non-identifiability is primarily a coefficient problem. The model may still make fitted predictions, but the individual coefficients cannot be interpreted uniquely.

8.15 Non-Identifiability Example: Celsius and Fahrenheit

If a model includes an intercept, Celsius, and Fahrenheit, then one column is an exact affine transformation of another:

$$F_i = 1.8C_i + 32. \quad (8.43)$$

In column form,

$$\mathbf{f} = 1.8\mathbf{c} + 32\mathbf{1}. \quad (8.44)$$

Thus the columns $\mathbf{1}$, $\mathbf{c}$, and $\mathbf{f}$ are linearly dependent. The fitted values may still be unique, but the individual Celsius and Fahrenheit coefficients are not.

8.16 Common Sources of Non-Identifiability

Exact non-identifiability usually comes from exact linear dependence among columns of the design matrix. Common examples include:

1. **Duplicate variables:** including the same predictor twice.
2. **Unit conversions:** including meters and feet, or Celsius and Fahrenheit, in a model with an intercept.
3. **The dummy variable trap:** including an intercept and all category indicator columns.
4. **Derived totals:** including all component variables and their exact sum.
5. **Too many predictors:** if $p + 1 > n$, then $\mathbf{X}$ cannot have full column rank.

Dummy variable trap. Suppose a categorical variable has three levels: Army, Navy, and Marine Corps. Let $\mathbf{d}_A, \mathbf{d}_N, \mathbf{d}_M$ be the three indicator columns. For every row,

$$d_{iA} + d_{iN} + d_{iM} = 1. \quad (8.45)$$

Thus,

$$\mathbf{d}_A + \mathbf{d}_N + \mathbf{d}_M = \mathbf{1}. \quad (8.46)$$

If the design matrix includes $\mathbf{1}$, $\mathbf{d}_A$, $\mathbf{d}_N$, and $\mathbf{d}_M$, then its columns are linearly dependent. The usual repair is to include the intercept and drop one category indicator as the reference group.

8.17 Near Non-Identifiability: Multicollinearity

Exact linear dependence causes exact non-identifiability. Near linear dependence causes instability. This is multicollinearity.

If one column of $\mathbf{X}$ is almost a linear combination of other columns, then $\mathbf{X}^T \mathbf{X}$ may still be invertible, but it is close to singular. The least squares coefficients exist, but small changes in the data can produce large changes in the coefficient estimates.

This matters because:

1. coefficient standard errors become large,

2. signs and magnitudes can be unstable,
3. individual coefficient interpretations become fragile, and
4. predictions may remain reasonable even while coefficients are hard to interpret.

Operational distinction.

Condition	Design-matrix geometry	Consequence
Exact dependence	Nonzero Null($\mathbf{X}$)	Coefficients not unique
Near dependence	Columns nearly dependent	Coefficients unstable
Full rank and well-conditioned	Columns independent and not nearly redundant	Coefficients more stable

8.18 Worked Example: Counting Regression Degrees of Freedom

If $n = 40$ and we fit an MLR model with an intercept and $p = 5$ predictors, then the full-rank design matrix has $p + 1 = 6$ columns. The fitted values live in a 6-dimensional subspace, and the residuals have

$$40 - 6 = 34 \quad (8.47)$$

residual degrees of freedom. The overall F -test has numerator degrees of freedom 5 and denominator degrees of freedom 34.

Coding companion reference. Package-specific Python/R commands for checking rank, residual degrees of freedom, and non-identifiability have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the degrees-of-freedom and subspace interpretation.

8.19 Common Mistakes

1. **Forgetting the intercept.** With an intercept and p predictors, the design matrix has $p + 1$ columns, not p .
2. **Mixing up fitted-space dimension and regression F -test numerator df.** The fitted subspace has dimension $p + 1$, but the overall F -test numerator df is p when comparing to the intercept-only model.
3. **Thinking residuals live in predictor space.** Residuals are n -vectors. They live in $\mathbb{R}^n$, not in $\mathbb{R}^p$.
4. **Thinking coefficient null-space directions are residual directions.** Null($\mathbf{X}$) lives in coefficient space, while Null($\mathbf{X}^T$) lives in observation space.
5. **Using $(\mathbf{X}^T \mathbf{X})^{-1}$ when $\mathbf{X}$ is rank deficient.** If columns are dependent, the inverse does not exist.
6. **Assuming fitted values are not unique when coefficients are not unique.** Coefficients may be non-unique while the projection $\hat{\mathbf{y}}$ is still unique.

7. **Checking only pairwise correlations for identifiability.** A column can be a combination of several others even if no single pair looks perfectly correlated.
8. **Including every dummy variable plus an intercept.** This creates exact linear dependence.
9. **Treating multicollinearity as a prediction failure only.** Multicollinearity often harms coefficient interpretation before it visibly harms prediction.

8.20 Chapter Summary

Degrees of freedom are leftover dimensions. An n -vector of observations begins with n independent coordinates. Estimating the mean consumes one dimension, leaving $n - 1$ dimensions for residual variation.

Regression generalizes the same idea. For a full-rank design matrix with an intercept and p predictors,

$$\text{df}_{\text{resid}} = n - p - 1. \quad (8.48)$$

The fitted values live in $\text{Col}(\mathbf{X})$, and the residuals live in $\text{Null}(\mathbf{X}^T)$.

Identifiability requires $\text{Null}(\mathbf{X}) = \{\mathbf{0}\}$. If the design matrix has dependent columns, multiple coefficient vectors can produce the same fitted values. Exact dependence creates non-identifiability; near dependence creates unstable coefficient estimates.

Can you do this? You should be able to count residual degrees of freedom, distinguish fitted-space dimension from regression-test numerator degrees of freedom, identify the four subspaces in regression language, and recognize common non-identifiability patterns such as duplicated variables, unit conversions, and the dummy-variable trap. The full reference text contains the complete Math Foundations source map and reusable study template.

Chapter 9

Multiple Linear Regression in Practice: Predictors, Hyperplanes, and the Advertising Example

9.1 The Big Picture

The previous chapters built the geometry of least squares. We now use that geometry in a practical multiple linear regression setting. The goal is to move from the algebra

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \tag{9.1}$$

to the workflow an analyst actually uses when fitting, interpreting, and defending a regression model.

The central message of this chapter is:

Multiple linear regression is the same projection problem as simple linear regression, but the fitted values now live in a higher-dimensional subspace determined by multiple predictors.

The practical challenge is that the model is still mathematically simple, but it becomes harder to visualize and harder to interpret casually. With one predictor, the least squares fit is a line. With two predictors, the fit is a plane. With three or more predictors, the fit is a hyperplane. The word changes, but the algebra stays the same.

The main questions in this chapter are:

1. Is at least one predictor useful in predicting the response?
2. Do all predictors help, or is only a subset useful?
3. How well does the model fit the data?
4. Given new predictor values, what response should we predict, and how accurate is that prediction?

We will use the `Advertising.csv` data as the running example. The response is `Sales`, and the predictors are advertising budgets for `TV`, `Radio`, and `Newspaper` across 200 markets.

We keep the Chapter 7 sample-level notation: $\mathbf{y}$ is the observed response vector for the current data set, $\mathbf{X}$ is the design matrix, and $\hat{\mathbf{y}}$ is the fitted vector.

Math Foundations Connection. This chapter uses vector notation, matrix dimensions, matrix-vector multiplication, column-space ideas, linear combinations, dot products, transposes, orthogonality, and rank/full-column-rank intuition. These are covered in the Math Foundations textbook, Chapters 2, 3, 5, 7, 10, 11, and 12.

9.2 Simple Linear Regression and Multiple Linear Regression Use the Same Model Form

In simple linear regression, there is one predictor. For observation i , the model is

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (9.2)$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (9.3)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}, \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}. \quad (9.4)$$

In multiple linear regression with p predictors, observation i has predictor values

$$x_{i1}, x_{i2}, \dots, x_{ip}. \quad (9.5)$$

The model is

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \epsilon_i. \quad (9.6)$$

In vector form, it is still

$$\boxed{\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}.} \quad (9.7)$$

The only change is the number of columns in $\mathbf{X}$:

$$\mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1p} \\ 1 & x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix} \in \mathbb{R}^{n \times (p+1)}. \quad (9.8)$$

The coefficient vector is

$$\boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_p \end{bmatrix} \in \mathbb{R}^{p+1}. \quad (9.9)$$

Key point. The notation $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ is not unique to multiple regression. It already describes simple linear regression. Multiple regression simply adds more predictor columns to the design matrix.

9.3 The Design Matrix View

The design matrix stores the predictors in a way that lets one matrix-vector multiplication generate all fitted values at once. For a coefficient vector $\boldsymbol{\beta}$,

$$\mathbf{X}\boldsymbol{\beta} = \begin{bmatrix} \beta_0 + \beta_1 x_{11} + \cdots + \beta_p x_{1p} \\ \beta_0 + \beta_1 x_{21} + \cdots + \beta_p x_{2p} \\ \vdots \\ \beta_0 + \beta_1 x_{n1} + \cdots + \beta_p x_{np} \end{bmatrix}. \quad (9.10)$$

Thus, the i th fitted value is the inner product of row i of $\mathbf{X}$ with $\boldsymbol{\beta}$:

$$\hat{y}_i = \mathbf{x}_{i,:}^T \hat{\boldsymbol{\beta}}. \quad (9.11)$$

There is an equally important column-space interpretation:

$$\hat{\mathbf{y}} = \hat{\beta}_0 \mathbf{1} + \hat{\beta}_1 \mathbf{x}_1 + \hat{\beta}_2 \mathbf{x}_2 + \cdots + \hat{\beta}_p \mathbf{x}_p. \quad (9.12)$$

Here $\mathbf{x}_j$ is the j th predictor column. Therefore,

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{X}). \quad (9.13)$$

Regression chooses the linear combination of design-matrix columns that gets closest to $\mathbf{y}$ in squared-error distance.

9.4 What Changes Geometrically? Lines, Planes, and Hyperplanes

A simple linear regression fit looks like a line in a two-dimensional scatterplot. With two predictors, the fitted relationship is a plane:

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2. \quad (9.14)$$

This plane lives in the ordinary three-dimensional plotting space with axes X_1 , X_2 , and Y .

With three predictors, the fitted relationship is

$$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X_1 + \hat{\beta}_2 X_2 + \hat{\beta}_3 X_3. \quad (9.15)$$

This is a hyperplane. It is not easy to draw because the graph would need one response axis plus three predictor axes.

With p predictors, the fitted relationship is

$$\boxed{\hat{Y} = \hat{\beta}_0 + \sum_{j=1}^p \hat{\beta}_j X_j.} \quad (9.16)$$

This is a p -dimensional hyperplane in $(p+1)$ -dimensional predictor-response plotting space.

Do not confuse two spaces. There are two geometric pictures in play:

1. The **predictor-response plot**: a line, plane, or hyperplane in the space with predictor axes and a response axis.
2. The **least-squares projection**: an orthogonal projection of $\mathbf{y} \in \mathbb{R}^n$ onto $\text{Col}(\mathbf{X}) \subseteq \mathbb{R}^n$.

Both are useful. The first helps us visualize the model. The second explains the algebra.

9.5 Least Squares Estimation in Multiple Regression

The fitted coefficient vector minimizes the residual sum of squares:

$$\hat{\boldsymbol{\beta}} = \arg \min_{\mathbf{b} \in \mathbb{R}^{p+1}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2. \quad (9.17)$$

The normal equations are

$$\mathbf{X}^T (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}. \quad (9.18)$$

Equivalently,

$$\mathbf{X}^T \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{X}^T \mathbf{y}. \quad (9.19)$$

If $\mathbf{X}$ has full column rank, then $\mathbf{X}^T \mathbf{X}$ is invertible, and

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}.} \quad (9.20)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y} = \mathbf{H}\mathbf{y}, \quad (9.21)$$

where

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (9.22)$$

is the hat matrix.

The residuals are

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I} - \mathbf{H})\mathbf{y}. \quad (9.23)$$

The residuals are orthogonal to every column of $\mathbf{X}$:

$$\boxed{\mathbf{X}^T \mathbf{e} = \mathbf{0}.} \quad (9.24)$$

This remains true whether there is one predictor or many.

9.6 The Advertising Data

The `Advertising.csv` data set contains observations from 200 markets. The selected response variable is

$$\text{Sales}, \quad (9.25)$$

measured in thousands of units sold. The predictors are advertising budgets, measured in thousands of dollars:

$$\text{TV}, \quad \text{Radio}, \quad \text{Newspaper}. \quad (9.26)$$

A natural multiple linear regression model is

$$\text{Sales}_i = \beta_0 + \beta_1 \text{TV}_i + \beta_2 \text{Radio}_i + \beta_3 \text{Newspaper}_i + \epsilon_i. \quad (9.27)$$

In vector form,

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (9.28)$$

where

$$\mathbf{X} = \begin{bmatrix} 1 & \text{TV}_1 & \text{Radio}_1 & \text{Newspaper}_1 \\ 1 & \text{TV}_2 & \text{Radio}_2 & \text{Newspaper}_2 \\ \vdots & \vdots & \vdots & \vdots \\ 1 & \text{TV}_{200} & \text{Radio}_{200} & \text{Newspaper}_{200} \end{bmatrix}. \quad (9.29)$$

Here

$$n = 200, \quad p = 3, \quad \mathbf{X} \in \mathbb{R}^{200 \times 4}. \quad (9.30)$$

If the columns are linearly independent, then the residual degrees of freedom are

$$n - p - 1 = 200 - 3 - 1 = 196. \quad (9.31)$$

9.7 A Scatterplot Matrix Is the First Reconnaissance Tool

With one predictor, a scatterplot can show the relationship between the predictor and response. With three predictors, we need to inspect several pairwise relationships. A scatterplot matrix displays the pairwise relationships among

$$\text{Sales}, \quad \text{TV}, \quad \text{Radio}, \quad \text{Newspaper}. \quad (9.32)$$

A scatterplot matrix helps answer two first-pass questions:

1. Which predictors appear most strongly associated with **Sales**?
2. Which predictors appear most strongly associated with each other?

The first question is about predictive signal. The second question is about collinearity. A predictor can look useful by itself but become less useful once another correlated predictor is included in the model.

Important caution. Pairwise plots are not the final analysis. They are reconnaissance. Multiple regression coefficients are interpreted after holding the other included predictors fixed. Pairwise plots do not automatically show that adjusted relationship.

9.8 Question 1: Is At Least One Predictor Useful?

The first model-level question is whether at least one predictor is useful in predicting the response. For the Advertising model,

$$\text{Sales}_i = \beta_0 + \beta_1 \text{TV}_i + \beta_2 \text{Radio}_i + \beta_3 \text{Newspaper}_i + \epsilon_i, \quad (9.33)$$

the null hypothesis is

$$H_0 : \beta_1 = \beta_2 = \beta_3 = 0. \quad (9.34)$$

The alternative is

$$H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \text{ is nonzero.} \quad (9.35)$$

The test statistic is the overall regression F -statistic:

$$F = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n - p - 1)}. \quad (9.36)$$

Here

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2, \quad (9.37)$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2. \quad (9.38)$$

For the Advertising full model, the numerator degrees of freedom are $p = 3$, and the denominator degrees of freedom are $n - p - 1 = 196$. A large F -statistic provides evidence that at least one advertising budget is associated with **Sales**.

9.9 Question 2: Which Predictors Help?

Once the global F -test suggests that at least one predictor is useful, the next question is which predictors matter after adjusting for the others.

For each coefficient β_j , the usual individual test is

$$H_0 : \beta_j = 0 \quad \text{versus} \quad H_A : \beta_j \neq 0. \quad (9.39)$$

The corresponding t -statistic has the form

$$t_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}. \quad (9.40)$$

In the full Advertising model, the fitted coefficient estimates are commonly reported approximately as

Term	Estimate	Std. error	t-statistic	p-value
Intercept	2.939	0.312	9.42	very small
TV	0.046	0.0014	32.81	very small
Radio	0.189	0.0086	21.89	very small
Newspaper	-0.001	0.0059	-0.18	0.86

These results suggest that **TV** and **Radio** are useful after adjusting for the other included predictors. **Newspaper** does not appear useful after adjusting for **TV** and **Radio**.

Interpretation of the coefficient for TV. Holding **Radio** and **Newspaper** fixed, an additional 1 thousand dollars in TV advertising is associated with about 0.046 thousand additional units sold. That is about 46 additional units sold.

Interpretation of the coefficient for Radio. Holding TV and **Newspaper** fixed, an additional 1 thousand dollars in **Radio** advertising is associated with about 0.189 thousand additional units sold. That is about 189 additional units sold.

Interpretation of the coefficient for Newspaper. Holding TV and **Radio** fixed, an additional 1 thousand dollars in **Newspaper** advertising is associated with an estimated change near zero. In this model, **Newspaper** does not add meaningful predictive information after accounting for the other two media.

9.10 Why a Predictor Can Look Useful Alone but Not in the Full Model

A simple linear regression of **Sales** on **Newspaper** can show a positive association. But the full multiple regression model can assign **Newspaper** a coefficient near zero.

This is not a contradiction. It is a lesson about adjustment.

A simple regression asks:

As **Newspaper** changes, how does **Sales** tend to change, ignoring other predictors?

A multiple regression coefficient asks:

As **Newspaper** changes, how does **Sales** tend to change while TV and **Radio** are held fixed?

If **Newspaper** budgets are correlated with **Radio** budgets, then **Newspaper** may appear useful by itself because it is partly acting as a surrogate for **Radio**. Once **Radio** is included directly, **Newspaper** may no longer have its own contribution.

Reusable diagnostic phrase. Simple regression measures marginal association. Multiple regression measures adjusted association.

9.11 Question 3: How Well Does the Model Fit?

Two common fit summaries are the residual standard error and R^2 .

The residual standard error is

$$\text{RSE} = \sqrt{\frac{\text{RSS}}{n - p - 1}}. \quad (9.41)$$

It measures a typical residual size in the response units. For the Advertising full model, RSE is about 1.69. Since **Sales** is measured in thousands of units, this corresponds to a typical residual size of about 1,690 units.

The coefficient of determination is

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} = \frac{\text{TSS} - \text{RSS}}{\text{TSS}}. \quad (9.42)$$

It measures the proportion of sample variation in **y** explained by the fitted model. For the Advertising full model, R^2 is about 0.897, meaning the model explains about 89.7 percent of the sample variation in **Sales**.

Important limitation. Training-set R^2 does not prove that the model will predict well on new data. It also does not prove that the model assumptions are correct. It is a fit summary, not a full validation report.

9.12 Adjusted R^2 as a Quick Complexity Penalty

Ordinary R^2 cannot decrease when a predictor is added to the model, because adding a column to **X** can only reduce or maintain the residual sum of squares. This can make ordinary R^2 too generous when comparing models with different numbers of predictors.

Adjusted R^2 introduces a degrees-of-freedom penalty:

$$R_{\text{adj}}^2 = 1 - \frac{\text{RSS}/(n - p - 1)}{\text{TSS}/(n - 1)}. \quad (9.43)$$

Unlike ordinary R^2 , adjusted R^2 can decrease when a predictor is added. It asks whether the reduction in RSS was large enough to justify the extra parameter.

Practical use. Adjusted R^2 is a quick model-comparison tool, but it should not replace out-of-sample validation, residual diagnostics, or domain reasoning.

9.13 Question 4: What Should We Predict and How Accurate Is the Prediction?

For new predictor values

$$\mathbf{x}_0 = [1 \quad x_{01} \quad x_{02} \quad \cdots \quad x_{0p}]^T, \quad (9.44)$$

the point prediction is

$$\boxed{\hat{y}_0 = \mathbf{x}_0^T \hat{\boldsymbol{\beta}}.} \quad (9.45)$$

For the Advertising model, suppose a market has

$$\text{TV} = 100, \quad \text{Radio} = 20, \quad \text{Newspaper} = 30, \quad (9.46)$$

where each value is in thousands of dollars. Using the approximate full-model estimates,

$$\widehat{\text{Sales}} \approx 2.939 + 0.046(100) + 0.189(20) - 0.001(30) \quad (9.47)$$

$$= 2.939 + 4.600 + 3.780 - 0.030 \quad (9.48)$$

$$= 11.289. \quad (9.49)$$

The predicted sales are about 11.289 thousand units, or 11,289 units.

There are two common uncertainty intervals:

1. A **confidence interval for the mean response**: uncertainty about the average sales among many markets with these same predictor values.
2. A **prediction interval for a new response**: uncertainty about the sales in one individual new market.

The prediction interval is wider because it includes both uncertainty in the estimated mean and the irreducible noise for a single future market.

9.14 Worked Example: Interpreting a Full Advertising Model

Suppose the fitted model is

$$\widehat{\text{Sales}} = \hat{\beta}_0 + \hat{\beta}_1 \text{TV} + \hat{\beta}_2 \text{Radio} + \hat{\beta}_3 \text{Newspaper}. \quad (9.50)$$

Each slope is interpreted while holding the other advertising budgets fixed. For example, $\hat{\beta}_1$ estimates the expected change in sales for a one-unit increase in the TV budget, with Radio and Newspaper held constant. If all three budgets change, the predicted change is the sum of the three coefficient-weighted changes. The full reference text gives the longer worked interpretation examples.

9.15 Assumptions Behind the Practical Regression Workflow

For estimating a useful least squares line or hyperplane, the weakest core condition is often written as

$$\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}. \quad (9.51)$$

This says that the included predictors capture the conditional mean structure, leaving errors with mean zero after conditioning on the design matrix.

For the usual standard errors, t -tests, and F -tests, a common working assumption is

$$\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}. \quad (9.52)$$

This says the errors have constant variance and are uncorrelated across observations.

For exact finite-sample normal-theory inference, we often further assume

$$\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}). \quad (9.53)$$

This assumption is stronger than what is needed for least squares estimation itself, but it supports exact t - and F -distribution calculations in the classical model.

9.16 Why Graphics Become Difficult in Higher Dimensions

In simple linear regression, one scatterplot can show the predictor, response, fitted line, and residual pattern. In multiple regression, one plot cannot show everything.

With two predictors, a 3D plot can show a regression plane, but even then it can be hard to see residual patterns, leverage, and interactions. With three or more predictors, no ordinary static plot can show the full fitted hyperplane.

Therefore, the practical workflow shifts toward:

1. scatterplot matrices for pairwise reconnaissance,
2. correlation matrices for predictor relationships,
3. coefficient tables for adjusted effects,
4. residual plots for assumption checks,
5. partial residual or component-plus-residual plots for predictor-specific patterns,
6. test-set or cross-validation performance for generalization.

9.17 A Practical Modeling Workflow for the Advertising Example

A defensible multiple linear regression workflow for the Advertising data could look like this:

1. State the response and predictor units.
2. Create scatterplots of **Sales** against each predictor.
3. Create a scatterplot matrix for **Sales**, **TV**, **Radio**, and **Newspaper**.
4. Check predictor-predictor relationships, especially possible collinearity.
5. Fit the full model with all three predictors.
6. Use the overall F -test to ask whether at least one predictor is useful.
7. Use coefficient tests and domain reasoning to ask which predictors contribute after adjustment.
8. Compare simpler models such as **Sales** on **TV** and **Radio** only.
9. Report RSE and R^2 , but do not stop there.
10. Check residual plots for nonlinearity, heteroskedasticity, and unusual observations.
11. Use prediction or confidence intervals depending on whether the question is about one future market or the mean response at a predictor setting.
12. Document all modeling decisions so the analysis is reproducible and defensible.

Coding companion reference. Package-specific Python/R commands for scatterplot matrices, full and reduced multiple-regression models, prediction intervals, and residual plots have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the applied modeling workflow and interpretation guidance.

9.18 Common Mistakes

1. **Forgetting that coefficients are adjusted effects.** In MLR, $\hat{\beta}_j$ is interpreted while holding the other included predictors fixed.
2. **Treating pairwise plots as final proof.** Scatterplot matrices are useful reconnaissance, but they do not replace adjusted model interpretation.
3. **Confusing units.** In the Advertising data, `Sales` is in thousands of units and budgets are in thousands of dollars.
4. **Saying a nonsignificant coefficient proves no relationship.** It means the fitted model did not provide strong evidence for an adjusted linear effect under the model assumptions.
5. **Ignoring correlation among predictors.** Predictor-predictor relationships can change coefficient interpretation and standard errors.
6. **Using ordinary R^2 alone to choose variables.** Ordinary R^2 increases or stays the same when predictors are added.
7. **Mixing confidence intervals and prediction intervals.** Confidence intervals are for mean response; prediction intervals are for individual future responses.
8. **Drawing the hyperplane but skipping residual checks.** A fitted plane can still be the wrong shape if the true relationship is nonlinear or contains interactions.
9. **Assuming predictive association is causal.** Regression on observational data can estimate associations, but causal claims require design, assumptions, or additional evidence.

9.19 Chapter Summary

Multiple linear regression uses the same least-squares projection idea as simple linear regression, but the design matrix has more predictor columns. With an intercept and p predictors,

$$\mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \hat{\mathbf{y}} = \mathbf{X}\hat{\boldsymbol{\beta}}. \quad (9.54)$$

The fitted values live in the column space of $\mathbf{X}$. With one predictor, the fit is a line; with two predictors, it is a plane; with more predictors, it is a hyperplane.

The practical MLR questions are the same ones emphasized in the Advertising example: whether at least one predictor is useful, which predictors help after controlling for the others, how well the model fits, and what predictions the model makes for new predictor values. The overall F -test addresses the first question; individual coefficient tests address one predictor at a time; R^2 , adjusted R^2 , RSE, and prediction intervals help describe fit and prediction quality.

Can you do this? You should be able to write the MLR design matrix, interpret a coefficient while holding other predictors fixed, distinguish the overall F -test from individual t -tests, explain why simple and multiple regression can disagree, and brief the main findings from an Advertising-style model. The full reference text contains the complete Math Foundations source map and reusable study template.

Chapter 10

Truth + Error, Estimator Behavior, and Residual Geometry

10.1 The Big Picture

The previous chapter treated multiple linear regression as a practical modeling tool. We used the design matrix, the coefficient vector, and the least squares fit to answer applied questions about predictors, model fit, and prediction. This chapter steps back into theory for one reason:

We want to understand what the least squares estimates and residuals are doing, so that later diagnostics and hypothesis tests are more than button-clicking.

The model is still

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (10.1)$$

The difference is that we now treat this equation as a *truth plus error* representation. The term $\mathbf{X}\boldsymbol{\beta}$ is the systematic part of the response, and $\boldsymbol{\epsilon}$ is the random part that the model does not explain.

Notation bridge. Chapters 7, 9, 11, and 12 use $\mathbf{y}$ for the observed response vector from a realized data set. This chapter temporarily switches to $\mathbf{Y}$ because the full response vector is being treated as random once the design matrix $\mathbf{X}$ is fixed.

The central questions are:

1. If $\boldsymbol{\beta}$ is the true coefficient vector, how does $\hat{\boldsymbol{\beta}}$ behave?
2. Is $\hat{\boldsymbol{\beta}}$ centered on the true value $\boldsymbol{\beta}$?
3. How variable is $\hat{\boldsymbol{\beta}}$?
4. How do the residuals $\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}$ relate to the true errors $\boldsymbol{\epsilon}$?
5. Why do residual plots tell us whether our modeling assumptions are plausible?

This chapter is the bridge from regression geometry to regression diagnostics. It explains why residuals are useful, but also why they are not the same thing as the true errors.

Math Foundations Connection. This chapter uses vector notation, transposes, matrix-vector multiplication, identity matrices, orthogonality, rank, column-space/residual-space geometry, least-squares normal equations, and the $\mathbf{A}^T \mathbf{A}$ Hessian. These connections are covered in the Math Foundations textbook, Chapters 2, 5, 7, 10, 11, 12, and 13.

10.2 The Regression Model with Fixed Design Matrix

We start with the multiple linear regression model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (10.2)$$

where

$$\mathbf{Y} \in \mathbb{R}^n, \quad \mathbf{X} \in \mathbb{R}^{n \times (p+1)}, \quad \boldsymbol{\beta} \in \mathbb{R}^{p+1}, \quad \boldsymbol{\epsilon} \in \mathbb{R}^n. \quad (10.3)$$

The design matrix $\mathbf{X}$ includes an intercept column of ones and p predictor columns.

In this chapter, we use a fixed-design view:

Treat $\mathbf{X}$ as fixed, and let $\mathbf{Y}$ vary randomly because $\boldsymbol{\epsilon}$ varies randomly.

That is the answer to the question, “Where does the randomness come from?” It comes from the noise vector $\boldsymbol{\epsilon}$. Once $\mathbf{X}$ is fixed, the only random object on the right side of (10.2) is $\boldsymbol{\epsilon}$, so the only random object on the left side is $\mathbf{Y}$.

Conditional view. If the predictors are themselves random in the data-generating process, the results in this chapter are usually read conditionally on the observed $\mathbf{X}$. In words: after we observe the predictor values, we analyze the randomness in the response around the regression surface.

10.3 The Big Three Regression Assumptions

Chapter 5 already introduced the shared assumption ladder. Here we restate it in matrix form.

1. Assumption 1: Mean-zero errors

$$\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}. \quad (10.4)$$

Then $\mathbb{E}[\mathbf{Y} \mid \mathbf{X}] = \mathbf{X}\boldsymbol{\beta}$.

2. Assumption 2: Constant variance and uncorrelated errors

$$\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n. \quad (10.5)$$

This is the matrix form of homoskedastic, uncorrelated errors.

3. Assumption 3: Gaussian errors

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n). \quad (10.6)$$

This supports exact classical t - and F -based inference.

The Core Reader uses this section as a recall, not a full re-teaching of the ladder.

10.4 Recall: Least Squares Estimator

The estimator itself is not new. From Chapter 7,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}, \quad (10.7)$$

assuming $\mathbf{X}$ has full column rank. The fitted response vector is

$$\hat{\mathbf{Y}} = \mathbf{X} \hat{\boldsymbol{\beta}}, \quad (10.8)$$

and the residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}. \quad (10.9)$$

The new point in this chapter is not the formula itself. It is the sampling behavior of that formula when $\mathbf{Y}$ is random under a fixed design matrix.

10.5 Dimension Check

The formulas in this chapter are much easier to trust if the dimensions work. Let $q = p + 1$, the number of columns in $\mathbf{X}$. Then:

Object	Meaning	Dimension
$\mathbf{Y}$	response vector	$n \times 1$
$\mathbf{X}$	design matrix	$n \times q$
$\boldsymbol{\beta}$	true coefficient vector	$q \times 1$
$\hat{\boldsymbol{\beta}}$	estimated coefficient vector	$q \times 1$
$\boldsymbol{\epsilon}$	true error vector	$n \times 1$
$\hat{\mathbf{Y}}$	fitted response vector	$n \times 1$
$\mathbf{e}$	residual vector	$n \times 1$
$\mathbf{H}$	hat matrix	$n \times n$
$\mathbf{I}_n - \mathbf{H}$	residual-maker matrix	$n \times n$

The matrix

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \quad (10.10)$$

has dimension $q \times n$. It maps the response vector $\mathbf{Y} \in \mathbb{R}^n$ into the coefficient estimate $\hat{\boldsymbol{\beta}} \in \mathbb{R}^q$:

$$\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{Y}. \quad (10.11)$$

10.6 Truth + Error Representation for $\hat{\boldsymbol{\beta}}$

Start with the least squares estimator:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{Y}. \quad (10.12)$$

Substitute the regression model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (10.13)$$

$$= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X}\boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon} \quad (10.14)$$

$$= \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}. \quad (10.15)$$

Therefore,

$$\boxed{\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon}.} \quad (10.16)$$

This is the truth plus error representation for the coefficient estimator. It says:

The estimated coefficient vector equals the true coefficient vector plus a transformed version of the random noise vector.

Each coefficient estimate has its own associated error term. That error is not a new source of randomness; it is produced by the same $\boldsymbol{\epsilon}$ that created the random response values.

10.7 Unbiasedness of $\hat{\boldsymbol{\beta}}$

Using (10.16), take conditional expectations given $\mathbf{X}$:

$$\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \mathbb{E}[\boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \boldsymbol{\epsilon} \mid \mathbf{X}] \quad (10.17)$$

$$= \boldsymbol{\beta} + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}]. \quad (10.18)$$

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\hat{\boldsymbol{\beta}} \mid \mathbf{X}] = \boldsymbol{\beta}.} \quad (10.19)$$

So $\hat{\boldsymbol{\beta}}$ is unbiased under the mean-zero error assumption.

What unbiased means here. Unbiased does not mean that $\hat{\boldsymbol{\beta}}$ equals $\boldsymbol{\beta}$ in a particular dataset. It means that if we repeatedly observed new response vectors from the same design matrix and same true model, the average of the coefficient estimates would center on the true coefficient vector.

10.8 Variance of $\hat{\boldsymbol{\beta}}$

The truth plus error representation also makes the variance calculation direct. Define

$$\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (10.20)$$

Then

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \mathbf{A}\boldsymbol{\epsilon}. \quad (10.21)$$

Since $\boldsymbol{\beta}$ is fixed, it has no variance. Therefore,

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \text{Var}(\mathbf{A}\boldsymbol{\epsilon} \mid \mathbf{X}). \quad (10.22)$$

The matrix variance rule is

$$\text{Var}(\mathbf{AZ}) = \mathbf{A} \text{Var}(\mathbf{Z}) \mathbf{A}^T. \quad (10.23)$$

Using $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$,

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \mathbf{A} \text{Var}(\epsilon \mid \mathbf{X}) \mathbf{A}^T \quad (10.24)$$

$$= \mathbf{A}(\sigma^2 \mathbf{I}_n) \mathbf{A}^T \quad (10.25)$$

$$= \sigma^2 \mathbf{A} \mathbf{A}^T. \quad (10.26)$$

Now substitute $\mathbf{A} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T$:

$$\mathbf{A} \mathbf{A}^T = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T [(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T]^T \quad (10.27)$$

$$= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{X} [(\mathbf{X}^T \mathbf{X})^{-1}]^T. \quad (10.28)$$

Because $\mathbf{X}^T \mathbf{X}$ is symmetric, its inverse is also symmetric. Therefore,

$$\mathbf{A} \mathbf{A}^T = (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.29)$$

So

$$\boxed{\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^T \mathbf{X})^{-1}.} \quad (10.30)$$

What this means. The coefficient uncertainty depends on two things:

1. the error variance σ^2 , and
2. the geometry of the design matrix through $(\mathbf{X}^T \mathbf{X})^{-1}$.

If the errors are noisy, coefficient estimates are noisy. If the predictor columns are poorly arranged, nearly dependent, or badly scaled, then $(\mathbf{X}^T \mathbf{X})^{-1}$ can contain large entries, making coefficient estimates unstable.

10.9 Standard Errors of the Coefficients

The diagonal entries of $\text{Var}(\hat{\beta} \mid \mathbf{X})$ give the variances of the individual coefficient estimates. Let

$$\mathbf{C} = (\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.31)$$

Then

$$\text{Var}(\hat{\beta}_j \mid \mathbf{X}) = \sigma^2 C_{jj}. \quad (10.32)$$

The standard error of $\hat{\beta}_j$ is

$$\text{SE}(\hat{\beta}_j) = \sigma \sqrt{C_{jj}}. \quad (10.33)$$

Because σ is unknown, we estimate it using the residuals:

$$\hat{\sigma}^2 = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1} = \frac{\text{RSS}}{n - p - 1}. \quad (10.34)$$

Thus the estimated standard error is

$$\boxed{\widehat{\text{SE}}(\hat{\beta}_j) = \hat{\sigma} \sqrt{C_{jj}}.} \quad (10.35)$$

This is the quantity reported in regression output tables. It is the denominator of the usual t -statistic.

10.10 Distribution of $\hat{\beta}$

If Gaussian errors are added to the first two assumptions, then $\hat{\beta}$ is a linear transformation of a multivariate normal vector, so

$$\hat{\beta} \mid \mathbf{X} \sim N(\beta, \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}). \quad (10.36)$$

This is the distributional step that justifies the usual coefficient tests and confidence intervals.

10.11 Takeaways for Coefficients

The coefficient estimator has three central properties under the standard linear model assumptions:

1. **Truth plus error:**

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon. \quad (10.37)$$

2. **Unbiasedness under mean-zero errors:**

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta. \quad (10.38)$$

3. **Variance under homoskedastic uncorrelated errors:**

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}. \quad (10.39)$$

If the Gaussian assumption is added, then $\hat{\beta}$ has a multivariate normal distribution.

10.12 The Hat Matrix Revisited

This chapter reuses the same projection matrix from Chapter 7:

$$\mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T, \quad (10.40)$$

so

$$\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}. \quad (10.41)$$

The main recall facts are unchanged: $\mathbf{H}$ is symmetric, idempotent, and projects onto $\text{Col}(\mathbf{X})$.

10.13 The Residual Vector

The residual vector is

$$\mathbf{e} = \mathbf{Y} - \hat{\mathbf{Y}}. \quad (10.42)$$

Using $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$,

$$\mathbf{e} = \mathbf{Y} - \mathbf{H}\mathbf{Y} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}. \quad (10.43)$$

The matrix

$$\mathbf{M} = \mathbf{I}_n - \mathbf{H} \quad (10.44)$$

is often called the residual-maker matrix because

$$\mathbf{e} = \mathbf{M}\mathbf{Y}. \quad (10.45)$$

Now substitute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) \quad (10.46)$$

$$= (\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} + (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.47)$$

Because $\mathbf{H}$ projects onto $\text{Col}(\mathbf{X})$, it leaves every vector in $\text{Col}(\mathbf{X})$ unchanged. Since $\mathbf{X}\boldsymbol{\beta} \in \text{Col}(\mathbf{X})$,

$$\mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}. \quad (10.48)$$

Therefore,

$$(\mathbf{I}_n - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}, \quad (10.49)$$

and

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \quad (10.50)$$

This is the truth plus error representation for residuals.

Important interpretation. Residuals are not the true errors. The true errors are $\boldsymbol{\epsilon}$. The residuals are projected errors:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.51)$$

Least squares removes from $\boldsymbol{\epsilon}$ the component that lies in $\text{Col}(\mathbf{X})$. What remains is the component orthogonal to the fitted-value space.

10.14 Why $\mathbf{I}_n - \mathbf{H}$ Is Also an Orthogonal Projection

Since $\mathbf{H}$ projects onto the fitted-value space, the complementary matrix

$$\mathbf{I}_n - \mathbf{H} \quad (10.52)$$

projects onto the residual space. It is also symmetric and idempotent, so it is an orthogonal projection matrix as well. That is why residuals live in the orthogonal complement of $\text{Col}(\mathbf{X})$.

10.15 Expectation of the Residual Vector

Using (10.50),

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.53)$$

Take conditional expectations:

$$\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = (\mathbf{I}_n - \mathbf{H})\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}]. \quad (10.54)$$

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{X}] = \mathbf{0}$, then

$$\boxed{\mathbb{E}[\mathbf{e} \mid \mathbf{X}] = \mathbf{0}.} \quad (10.55)$$

This is why we expect residual plots to be centered around zero when the mean structure is correctly specified.

10.16 Variance of the Residual Vector

Again use

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}. \quad (10.56)$$

By the matrix variance rule,

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = (\mathbf{I}_n - \mathbf{H}) \text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X})(\mathbf{I}_n - \mathbf{H})^T. \quad (10.57)$$

If $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2 (\mathbf{I}_n - \mathbf{H})(\mathbf{I}_n - \mathbf{H})^T \quad (10.58)$$

$$= \sigma^2 (\mathbf{I}_n - \mathbf{H})^2 \quad (10.59)$$

$$= \sigma^2 (\mathbf{I}_n - \mathbf{H}). \quad (10.60)$$

Thus

$$\boxed{\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2 (\mathbf{I}_n - \mathbf{H}).} \quad (10.61)$$

This formula has an important consequence. The i th residual has variance

$$\text{Var}(e_i \mid \mathbf{X}) = \sigma^2 (1 - h_{ii}), \quad (10.62)$$

where h_{ii} is the i th diagonal entry of the hat matrix. The off-diagonal entries give residual covariances:

$$\text{Cov}(e_i, e_j \mid \mathbf{X}) = -\sigma^2 h_{ij}, \quad i \neq j. \quad (10.63)$$

Key diagnostic warning. Even if the true errors are uncorrelated and have equal variance, the raw residuals generally do not have equal variance and are generally correlated. This is why leverage-adjusted residuals and studentized residuals are useful in diagnostics.

10.17 Distribution of the Residual Vector

If the Gaussian assumption holds, then the residual vector is also multivariate normal:

$$\mathbf{e} \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 (\mathbf{I}_n - \mathbf{H})). \quad (10.64)$$

Its covariance matrix is singular because residuals are constrained to the residual subspace rather than all of $\mathbb{R}^n$.

10.18 Residual Degrees of Freedom

The residual degrees of freedom are the dimensions left over after fitting the model. Algebraically,

$$\text{df}_{\text{res}} = \text{tr}(\mathbf{I}_n - \mathbf{H}). \quad (10.65)$$

Because $\text{tr}(\mathbf{I}_n) = n$ and $\text{tr}(\mathbf{H}) = p + 1$ when $\mathbf{X}$ has full column rank,

$$\boxed{\text{df}_{\text{res}} = n - p - 1.} \quad (10.66)$$

This also explains the denominator in the residual variance estimate:

$$\hat{\sigma}^2 = \frac{\mathbf{e}^T \mathbf{e}}{n - p - 1}. \quad (10.67)$$

The numerator $\mathbf{e}^T \mathbf{e}$ is the residual sum of squares. The denominator is the number of leftover residual dimensions.

10.19 What Residual Plots Are Really Checking

The residual representation

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon} \quad (10.68)$$

explains why residual analysis works. If the true errors have mean zero, constant variance, and approximate normality, then the residuals should show compatible behavior.

Common diagnostic checks include:

Diagnostic plot	What we hope to see	Possible problem if violated
Residuals vs fitted values	random cloud around zero	nonlinearity or missing structure
Residuals vs predictor	random cloud around zero	predictor transformation needed
Scale-location plot	similar vertical spread	heteroskedasticity
Q-Q plot	points close to a line	non-normal error behavior
Residuals vs time/order	no runs or waves	correlated errors
Residuals vs leverage	no extreme influential points	high leverage or outliers

A good residual plot does not prove that the model is true. It only suggests that the most visible violations are not obvious. A bad residual plot is more informative: it gives evidence that something in the modeling assumptions or model form may need attention.

10.20 Good Residual Behavior

A residual plot with no visible pattern is usually a good sign. A healthy residual-vs-fitted plot should look like a random scatter of points around the horizontal line at zero. The points do not need to be perfectly symmetric, but they should not show a systematic curve, trend, funnel, or clustering pattern.

A random-looking residual cloud suggests:

1. the conditional mean may be reasonably specified,
2. the variance may be roughly stable over the fitted range, and
3. there may not be an obvious missing nonlinear term.

This is why residuals are the natural next object after coefficients. Coefficients tell us what the fitted model says. Residuals tell us what the fitted model failed to explain.

10.21 Bad Residual Behavior

Bad residual behavior usually points to one of a few issues:

1. **Curvature:** residuals form a U-shape or other smooth pattern. This suggests the mean function is not linear in the current predictors.
2. **Funnel shape:** residual spread increases or decreases with fitted values. This suggests nonconstant variance.
3. **Runs over time:** residuals stay positive or negative for long stretches in time order. This suggests correlated errors.
4. **Extreme residuals:** a few residuals are much larger than the rest. This suggests outliers or observations not explained by the model.
5. **High leverage:** some rows of $\mathbf{X}$ are far from the rest. These observations may strongly influence the fitted model.

Residual analysis is not just a plot-making exercise. It is a theory-backed health check of the assumptions that allowed us to interpret $\hat{\beta}$, standard errors, t -statistics, and prediction intervals.

10.22 Residuals, Leverage, and Studentization

The diagonal hat values

$$h_{ii} \tag{10.69}$$

measure leverage. An observation has high leverage when its predictor pattern is unusual relative to the rest of the design matrix.

Because

$$\text{Var}(e_i | \mathbf{X}) = \sigma^2(1 - h_{ii}), \tag{10.70}$$

raw residuals do not all have the same variance. A leverage-adjusted residual divides by its estimated standard deviation:

$$r_i = \frac{e_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}. \tag{10.71}$$

This is often called an internally studentized residual. It gives a more comparable scale for judging which residuals are unusually large.

Why leverage matters. A point with a large residual but low leverage may be an outlier in the response direction. A point with high leverage may control the fitted hyperplane even if its residual is not large. A point with both high leverage and a large residual is especially important to inspect.

10.23 A Clean Interpretation of Diagnostics

The following table connects each theoretical assumption to the residual behavior we expect to see.

Assumption about ϵ	Residual implication	Diagnostic clue
$\mathbb{E}[\epsilon \mathbf{X}] = 0$	residuals centered near zero	no systematic trend
$\text{Var}(\epsilon \mathbf{X}) = \sigma^2 \mathbf{I}$	stable spread, no strong correlation	no funnel or tracking
Gaussian errors	residuals roughly normal after adjustment	Q-Q plot roughly linear
Correct linear mean structure	no missing pattern in residuals	no curve or clusters
Full column rank	unique coefficients	no exact multicollinearity

10.24 Why This Matters for Modeling Decisions

The coefficient theory and residual theory combine into a modeling workflow:

1. Fit a model.
2. Interpret $\hat{\beta}$, standard errors, and model fit.
3. Check residual behavior.
4. If residuals behave badly, consider model repair.

Model repair might include transforming predictors, transforming the response, adding interactions, adding nonlinear terms, using weighted least squares, using robust standard errors, or moving to a different model class. The next chapters build toward these tools.

The key point is that residual diagnostics are not separate from the math. They come directly from

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon \quad (10.72)$$

and

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.73)$$

Coding companion reference. Package-specific Python/R commands for extracting fitted values, residuals, leverage, studentized residuals, diagnostic plots, and coefficient covariance matrices have been moved to the separate Coding and Software Cookbook companion. The main chapter keeps the theoretical residual and estimator behavior.

10.25 Common Mistakes

Mistake 1: Treating residuals as the true errors

The residuals are not ϵ . They are

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon. \quad (10.74)$$

They are the part of the error vector left after projecting away the fitted-value component.

Mistake 2: Forgetting the fixed-design condition

The cleanest formulas are conditional on $\mathbf{X}$. If the predictors are random, we usually interpret the results after conditioning on the observed design matrix.

Mistake 3: Thinking unbiased means correct in this dataset

Unbiased means centered correctly over repeated samples. It does not mean the estimate is close in one observed dataset.

Mistake 4: Ignoring multicollinearity in the variance formula

The variance formula

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1} \quad (10.75)$$

shows why nearly dependent predictor columns can cause large standard errors.

Mistake 5: Using residual plots as proof

A clean residual plot is not proof that assumptions hold. It is evidence that obvious violations are not visible. A bad residual plot is stronger evidence because it points to a concrete failure mode.

10.26 Operational Briefing Checklist

For the Core Reader, condense the briefing to the essential questions:

1. What is the design matrix dimension and is $\mathbf{X}$ full column rank?
2. Which assumption level is needed for the claim being made?
3. What do the residual plots say about centering, spread, and pattern?
4. Are leverage or unusually large adjusted residuals driving the fit?
5. What repair move would follow if a diagnostic problem appears?

10.27 Chapter Summary

The core equations of this chapter are:

$$\hat{\beta} = \beta + (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \epsilon, \quad (10.76)$$

$$\mathbb{E}[\hat{\beta} \mid \mathbf{X}] = \beta, \quad (10.77)$$

$$\text{Var}(\hat{\beta} \mid \mathbf{X}) = \sigma^2(\mathbf{X}^T \mathbf{X})^{-1}, \quad (10.78)$$

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\epsilon, \quad (10.79)$$

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}). \quad (10.80)$$

The coefficient estimator is the truth plus a transformed error. Under mean-zero errors, it is unbiased. Under constant-variance uncorrelated errors, its variance is governed by $\sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$. Under Gaussian errors, it has a multivariate normal distribution.

The residual vector is the true error vector after projection onto the residual subspace. This explains why residuals can be used to diagnose model assumptions. Residuals should be centered around zero, have roughly stable spread, and show no systematic patterns if the model assumptions are reasonable. Chapter 12 turns these facts into a practical diagnostic and repair workflow.

10.28 Math Foundations Source Map

The Core Reader keeps the theory links to Math Foundations but omits the full Math Foundations textbook-location map. The main supporting ideas here are vectors, orthogonality, matrix multiplication, full column rank, and least-squares normal-equation structure.

10.29 Practice and Workflow

Can You Do This?

For Core Reader study, be sure you can do the following.

1. Write the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ and identify which object is random.
2. State the truth-plus-error forms for $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$.
3. Give the variance formulas for $\hat{\boldsymbol{\beta}}$ and $\mathbf{e}$.
4. Explain why residuals diagnose assumption failures instead of proving assumptions true.
5. Link a bad residual pattern to a likely repair move.

Chapter 11

Linear Models Beyond Straight Lines: Transformations and Linear-in-Parameters Models

11.1 The Big Picture

The previous chapters built multiple linear regression from two viewpoints:

1. Algebraically, the fitted values are produced by a design matrix and a coefficient vector.
2. Geometrically, the fitted response vector is the projection of $\mathbf{y}$ onto the column space of the design matrix.

This chapter makes an important expansion. Earlier chapters used "linear" mainly through the geometric picture of lines, planes, and hyperplanes in the displayed predictors. That picture was useful, but incomplete:

A linear model does not require the response to be a straight-line function of the original predictor values. It only requires the model to be linear in the unknown coefficients.

That distinction is easy to miss. The model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon \quad (11.1)$$

is curved as a function of x , but it is still a linear model in the parameters $\beta_0, \beta_1, \beta_2$. Once we create the transformed predictor column x^2 , ordinary least squares can be used exactly as before.

The central question is therefore not, "Is the graph a straight line?" The better question is:

Can the model be written as a design matrix times a coefficient vector, plus error?

If the answer is yes, then the linear regression machinery still applies.

We also return to the sample-level notation of Chapters 7 and 9: $\mathbf{y}$ is the observed response vector from the data set being fit.

Math Foundations Connection. This chapter uses the Math Foundations ideas of vectors, linear combinations, bases, matrix-vector multiplication, submatrices, linear independence, Gram-Schmidt orthogonalization, solving linear systems, and full-column-rank inverse-like formulas. In particular, transformed regression is just a new way of building columns for the design matrix. Those columns form candidate directions in $\mathbb{R}^n$, and least squares chooses the best linear combination of those directions. See the Math Foundations textbook, Chapters 2, 3, 5, 6, 7, 8, 9, 10, and 12.

11.2 What “Linear” Means Here

A model is linear in parameters when it can be written as

$$m(x; \boldsymbol{\theta}) = \theta_0 g_0(x) + \theta_1 g_1(x) + \cdots + \theta_q g_q(x), \quad (11.2)$$

where the functions $g_0, g_1, \dots, g_q$ are known functions of the input values, and the unknowns are the coefficients $\theta_0, \theta_1, \dots, \theta_q$. Usually $g_0(x) = 1$, which gives the intercept column.

The key is that the unknown parameters appear only as coefficients multiplying known columns. They are not multiplied by each other, raised to powers, placed inside nonlinear functions, or used as exponents.

For observations $i = 1, \dots, n$, define

$$z_{ij} = g_j(x_i), \quad j = 0, 1, \dots, q. \quad (11.3)$$

Then the model becomes

$$Y_i = \theta_0 z_{i0} + \theta_1 z_{i1} + \cdots + \theta_q z_{iq} + \epsilon_i. \quad (11.4)$$

Stacking the observations gives

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad (11.5)$$

where

$$\mathbf{Z} = \begin{bmatrix} z_{10} & z_{11} & \cdots & z_{1q} \\ z_{20} & z_{21} & \cdots & z_{2q} \\ \vdots & \vdots & \ddots & \vdots \\ z_{n0} & z_{n1} & \cdots & z_{nq} \end{bmatrix}, \quad \boldsymbol{\theta} = \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_q \end{bmatrix}. \quad (11.6)$$

This is the same matrix form as ordinary multiple linear regression. The only difference is that the design matrix may contain transformed versions of the original variables.

Original predictor space versus feature space

It is helpful to distinguish two spaces:

1. **Original predictor space:** the variables as measured, such as x , x_1 , x_2 , temperature, distance, or time.
2. **Feature space:** the columns actually used in the design matrix, such as x , x^2 , $\log x$, $\sin x$, $x_1 x_2$, or category indicators.

Linear regression is linear in the columns of the design matrix. Those columns can be created from nonlinear transformations of the original predictors.

11.3 The Least Squares Problem with Transformed Columns

Once the transformed design matrix $\mathbf{Z}$ is built, the least squares objective is

$$\min_{\mathbf{b} \in \mathbb{R}^{q+1}} \|\mathbf{y} - \mathbf{Z}\mathbf{b}\|^2. \quad (11.7)$$

The normal equations are

$$\mathbf{Z}^T \mathbf{Z} \hat{\boldsymbol{\theta}} = \mathbf{Z}^T \mathbf{y}. \quad (11.8)$$

If $\mathbf{Z}$ has full column rank, then

$$\boxed{\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}.} \quad (11.9)$$

The fitted values are

$$\hat{\mathbf{y}} = \mathbf{Z} \hat{\boldsymbol{\theta}}. \quad (11.10)$$

The hat matrix for the transformed model is

$$\mathbf{H}_{\mathbf{Z}} = \mathbf{Z}(\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T, \quad (11.11)$$

and therefore

$$\hat{\mathbf{y}} = \mathbf{H}_{\mathbf{Z}} \mathbf{y}, \quad \mathbf{e} = (\mathbf{I}_n - \mathbf{H}_{\mathbf{Z}}) \mathbf{y}. \quad (11.12)$$

Thus the geometry from Chapter 7 and the residual theory from Chapter 10 still apply. The projection subspace has changed because the design matrix columns changed, but the projection logic is the same.

11.4 A Dimension Check

If the transformed design matrix has $q + 1$ columns, then

$$\mathbf{Z} \in \mathbb{R}^{n \times (q+1)}, \quad \boldsymbol{\theta} \in \mathbb{R}^{q+1}. \quad (11.13)$$

The transformed model is identifiable when the columns of $\mathbf{Z}$ contain enough independent information to estimate $\boldsymbol{\theta}$ uniquely.

11.5 Mayer's Moon Motion Problem

Tobias Mayer's Moon-motion example motivates the chapter's central idea. The model used sine and cosine terms as predictors:

$$Y = \beta + \alpha x_1 + \gamma x_2 + \epsilon. \quad (11.14)$$

Even though x_1 and x_2 came from trigonometric transformations, the unknown coefficients still enter linearly. The model is therefore a linear model in the parameters. The full reference text gives the historical setup and geometry in more detail.

11.6 Basis Functions

A basis function is a known function used to build a column of the design matrix. In the notation from (11.2), the functions

$$g_0(x), g_1(x), \dots, g_q(x) \quad (11.15)$$

are basis functions.

A model using basis functions has the form

$$f(x) = \theta_0 g_0(x) + \theta_1 g_1(x) + \dots + \theta_q g_q(x). \quad (11.16)$$

This is a linear combination of known functions. The coefficients determine how much of each basis direction is used.

Common basis choices include:

Basis type	Example columns	Useful for
Polynomial	$1, x, x^2, x^3$	smooth curvature
Interaction	$1, x_1, x_2, x_1 x_2$	combined predictor effects
Trigonometric	$1, \sin x, \cos x$	cyclic or angular behavior
Logarithmic	$1, \log x$	diminishing returns and scale effects
Indicator/step	$I(x < c), I(x \geq c)$	piecewise constant effects
Spline	piecewise polynomial basis columns	flexible smooth curves

The design choice is not automatic. The analyst chooses basis functions based on domain knowledge, EDA, residual behavior, predictive performance, and interpretability.

11.7 Polynomial Regression

A polynomial regression model of degree d is

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \epsilon_i. \quad (11.17)$$

Although this model is nonlinear in x , it is linear in the coefficients $\beta_0, \beta_1, \dots, \beta_d$.

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 & \cdots & x_1^d \\ 1 & x_2 & x_2^2 & \cdots & x_2^d \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & x_n & x_n^2 & \cdots & x_n^d \end{bmatrix}. \quad (11.18)$$

This is a Vandermonde-style design matrix. The Math Foundations textbook uses a square Vandermonde matrix to fit an interpolating polynomial through n points; see Ch08 Sec. 8.7. In data analysis, we usually do something more modest: choose a small degree d , keep $d + 1 \ll n$, and solve a least squares problem instead of forcing the curve through every point.

Interpolation versus regression

Polynomial interpolation and polynomial regression can look similar algebraically, but their goals differ.

Task	Matrix shape	Goal
Interpolation	usually square $n \times n$	pass exactly through all points
Regression	usually tall $n \times (d + 1)$	find a stable trend with residual error

Interpolation can be useful in numerical methods, but in noisy data analysis it often overfits. Regression accepts residuals because the data include noise and because the goal is usually inference, prediction, or explanation rather than exact passage through every observed point.

Degree choice

A degree-one polynomial is the ordinary straight-line model. A degree-two polynomial allows one bend. A degree-three polynomial allows more shape. But high-degree polynomials can behave badly, especially near the edges of the observed data range.

A practical rule:

Use the lowest-degree polynomial that addresses the visible pattern and is defensible for the problem.

Residual plots and test error help decide whether the added flexibility is helping or merely fitting noise.

11.8 Interactions as Transformed Columns

Interactions are another example of linear-in-parameters modeling. With two predictors, the model

$$Y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i1} x_{i2} + \epsilon_i \quad (11.19)$$

is linear in the coefficients. The interaction column is simply

$$z_{i3} = x_{i1} x_{i2}. \quad (11.20)$$

The design matrix is

$$\mathbf{Z} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{11}x_{12} \\ 1 & x_{21} & x_{22} & x_{21}x_{22} \\ \vdots & \vdots & \vdots & \vdots \\ 1 & x_{n1} & x_{n2} & x_{n1}x_{n2} \end{bmatrix}. \quad (11.21)$$

The model can be rewritten as

$$Y_i = \beta_0 + (\beta_1 + \beta_3 x_{i2}) x_{i1} + \beta_2 x_{i2} + \epsilon_i. \quad (11.22)$$

Thus the slope with respect to x_1 depends on x_2 . This is why interactions are useful: they allow one predictor's effect to change depending on the value of another predictor, while preserving a linear-in-parameters model.

Hierarchy principle. If a model includes x_1x_2 , it is usually good practice to include x_1 and x_2 as well. This makes interpretation more stable and avoids treating the combined effect as meaningful while excluding its component directions.

11.9 Transforming the Response

So far, the transformations have been applied to predictors. Sometimes the response is transformed instead. Examples include

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (11.23)$$

or

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i. \quad (11.24)$$

A response transformation changes the scale on which the model is linear. For example, if

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (11.25)$$

then the model says that the conditional mean structure is simpler on a log scale than on the original scale. This can help when residual spread grows with the fitted values or when effects are multiplicative rather than additive.

Important caution. A transformed-response model is not the same as an untransformed-response model. If we fit $\log(Y)$, then predictions are naturally produced on the log scale. Transforming back with $\exp(\cdot)$ gives a prediction on the original scale, but uncertainty and bias require care because

$$\mathbb{E}[\exp(W)] \neq \exp(\mathbb{E}[W]) \quad (11.26)$$

in general.

11.10 Classifying Models: Linear or Not Linear?

The quickest check is to ask whether the model can be written as $\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}$ for a design matrix $\mathbf{Z}$ that can be computed from the observed data without knowing the parameters.

Model expression	Linear model?	Why
$\beta_0 + \beta_1 x$	Yes	coefficients multiply known columns
$\beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3$	Yes	polynomial columns are known
$\beta_0 x^{\beta_1}$	No	parameter appears as exponent
$\beta_0 + \beta_1 x_1 + \beta_0 \beta_1 x_1^2$	No	parameters are multiplied together
$1/(\beta_0 + \beta_1 x)$	No	parameters are inside reciprocal transformation
$\beta_1 \sin x + \beta_2 \cos x + \beta_3 \sin x \cos x$	Yes	transformed columns are known
$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2$	Yes	interaction column is known

The logistic regression mean function

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}} \quad (11.27)$$

is not a linear regression model for $p(x)$. It is, however, linear in the logit scale:

$$\log \left(\frac{p(x)}{1 - p(x)} \right) = \beta_0 + \beta_1 x. \quad (11.28)$$

That is why logistic regression belongs to the generalized linear model family rather than ordinary least squares linear regression.

11.11 Projection Geometry Still Works

A transformed design matrix changes the subspace, not the projection rule. In ordinary MLR, fitted values live in $\text{Col}(\mathbf{X})$. In this chapter, fitted values live in

$$\text{Col}(\mathbf{Z}). \quad (11.29)$$

Least squares chooses

$$\hat{\mathbf{y}} \in \text{Col}(\mathbf{Z}) \quad (11.30)$$

so that

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} \quad (11.31)$$

is orthogonal to every column of $\mathbf{Z}$. Therefore,

$$\mathbf{Z}^T \mathbf{e} = \mathbf{0}. \quad (11.32)$$

This is the transformed-model version of the normal equations.

Residual interpretation. If a residual plot shows curvature after fitting a straight-line model, we can add a transformed column that represents the missing shape. If the new transformed model is appropriate, the residual pattern should weaken or disappear. This does not prove the model is true, but it provides evidence that the new basis is capturing structure the old basis missed.

11.12 Truth Plus Error with Transformed Designs

All of the estimator behavior from the previous chapter carries over with $\mathbf{X}$ replaced by $\mathbf{Z}$. For the theoretical expectation and variance calculations, treat the response vector as random. If

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}, \quad (11.33)$$

then

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{Y} \quad (11.34)$$

$$= (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T (\mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}) \quad (11.35)$$

$$= \boldsymbol{\theta} + (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \boldsymbol{\epsilon}. \quad (11.36)$$

Thus,

$$\boxed{\hat{\boldsymbol{\theta}} = \boldsymbol{\theta} + (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \boldsymbol{\epsilon}.} \quad (11.37)$$

For the realized data set, we compute the coefficient estimate using the observed response vector:

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}. \quad (11.38)$$

The truth-plus-error expression above is used to study the estimator's behavior; it is not a computational formula because $\boldsymbol{\theta}$ and $\boldsymbol{\epsilon}$ are unknown.

If $\mathbb{E}[\boldsymbol{\epsilon} \mid \mathbf{Z}] = \mathbf{0}$, then

$$\mathbb{E}[\hat{\boldsymbol{\theta}} \mid \mathbf{Z}] = \boldsymbol{\theta}. \quad (11.39)$$

If $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{Z}) = \sigma^2 \mathbf{I}_n$, then

$$\text{Var}(\hat{\boldsymbol{\theta}} \mid \mathbf{Z}) = \sigma^2 (\mathbf{Z}^T \mathbf{Z})^{-1}. \quad (11.40)$$

This matters because transformed columns do not exempt us from regression assumptions. They only change the systematic part of the model.

11.13 Model Comparison for Transformations

When comparing candidate transformations, avoid relying on a single number or a single test in isolation. Useful evidence includes:

1. residual plots,
2. test MSE or cross-validation error,
3. adjusted R^2 , AIC, or BIC,
4. nested-model F -tests when the models are nested,
5. coefficient interpretability,
6. domain plausibility.

Nested transformed models

Suppose a reduced model has residual sum of squares RSS_R and degrees of freedom df_R , and a full model has residual sum of squares RSS_F and degrees of freedom df_F . If the reduced model is nested inside the full model, the usual comparison statistic is

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \quad (11.41)$$

For example, the straight-line model

$$Y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad (11.42)$$

is nested inside the quadratic model

$$Y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \epsilon_i. \quad (11.43)$$

The nested test asks whether the added x^2 column explains enough additional variation to justify the extra parameter.

Prediction-focused comparison

For prediction, a transformation that improves training RSS may not improve performance on new data. Use a train/test split or cross-validation when the modeling goal is predictive accuracy. This connects directly to the bias-variance ideas from earlier chapters: more basis columns increase flexibility, which can reduce bias but increase variance.

11.14 Practical Transformation Patterns

Polynomial terms

Use polynomial terms when residual plots suggest smooth curvature. Example:

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon. \quad (11.44)$$

Interpretation is about the whole curve, not just one coefficient in isolation.

Log predictor

Use $\log x$ when the effect of x appears to show diminishing returns or when x spans orders of magnitude:

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon. \quad (11.45)$$

This requires $x > 0$.

Log response

Use $\log Y$ when the response is positive and variability grows with the mean:

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon. \quad (11.46)$$

This requires $Y > 0$.

Square-root response

Use $\sqrt{Y}$ for nonnegative responses, especially count-like responses where variance grows with the mean:

$$\sqrt{Y} = \beta_0 + \beta_1 x + \epsilon. \quad (11.47)$$

This requires $Y \geq 0$.

Trigonometric terms

Use sine and cosine terms for cyclic predictors such as time of day, heading, angle, season, or orbital position:

$$Y = \beta_0 + \beta_1 \sin(t) + \beta_2 \cos(t) + \epsilon. \quad (11.48)$$

For a period P , use

$$\sin\left(\frac{2\pi t}{P}\right), \quad \cos\left(\frac{2\pi t}{P}\right). \quad (11.49)$$

Interactions

Use interactions when the effect of one predictor plausibly depends on another:

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \quad (11.50)$$

11.15 Numerical Stability and Identifiability

Transformations can create useful model flexibility, but they can also create columns that are redundant or nearly redundant. The two core checks are:

1. Does the transformed design matrix have full column rank?
2. Are some transformed columns nearly linear combinations of others?

Centering before creating powers can improve numerical stability, and orthogonal polynomial bases can reduce dependence among polynomial columns. The full reference text develops these ideas in more detail.

11.16 Interpretation After Transformations

Transformations improve flexibility, but they change interpretation.

Polynomial coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \epsilon, \quad (11.51)$$

the coefficient β_1 is not simply the global slope. The instantaneous slope of the fitted mean is

$$\frac{d}{dx} (\beta_0 + \beta_1 x + \beta_2 x^2) = \beta_1 + 2\beta_2 x. \quad (11.52)$$

Thus the effect of changing x depends on where you are on the curve.

Interaction coefficient interpretation

In

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon, \quad (11.53)$$

the slope with respect to x_1 is

$$\frac{\partial}{\partial x_1} \mathbb{E}[Y \mid x_1, x_2] = \beta_1 + \beta_3 x_2. \quad (11.54)$$

The coefficient β_3 tells us how the slope in x_1 changes as x_2 increases.

Log predictor interpretation

In

$$Y = \beta_0 + \beta_1 \log(x) + \epsilon, \quad (11.55)$$

small percentage changes in x correspond approximately to additive changes in Y . The exact change from x to cx is

$$\Delta \hat{Y} = \beta_1 \log(c). \quad (11.56)$$

Log response interpretation

In

$$\log(Y) = \beta_0 + \beta_1 x + \epsilon, \quad (11.57)$$

a one-unit increase in x changes the fitted log-response by β_1 . On the original scale, this corresponds approximately to multiplying the response by e^{β_1} , before accounting for retransformation issues.

11.17 Worked Example: Building a Polynomial Design Matrix

For a quadratic model,

$$y_i = \theta_0 + \theta_1 x_i + \theta_2 x_i^2 + \epsilon_i, \quad (11.58)$$

we build the transformed design matrix by adding a column of squared predictor values:

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}. \quad (11.59)$$

Once $\mathbf{Z}$ is built, least squares proceeds exactly as before:

$$\hat{\boldsymbol{\theta}} = (\mathbf{Z}^T \mathbf{Z})^{-1} \mathbf{Z}^T \mathbf{y}, \quad (11.60)$$

provided $\mathbf{Z}$ has full column rank.

11.18 Chapter Summary

A model can be curved in the original predictor values and still be a linear model. The key requirement is linearity in the unknown coefficients. After building transformed columns, the model has the same design-matrix form:

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \quad (11.61)$$

Polynomial terms, interactions, log terms, square-root terms, trigonometric terms, and indicator columns are all ways to build richer design matrices. The least-squares machinery still works as long as the transformed columns are well-defined and the design matrix has enough independent information.

The cost of transformations is interpretation and overfitting risk. A more flexible design can fit more patterns, but it can also chase noise, create unstable coefficients, or violate domain constraints.

Can you do this? You should be able to decide whether a model is linear in parameters, construct a transformed design matrix, interpret coefficients after transformations cautiously, and compare transformed models using residual behavior and validation performance. The full reference text contains the common-mistakes list, operational briefing checklist, complete Math Foundations source map, and reusable study template.

Chapter 12

Residual Analysis and Model Repair

12.1 The Big Picture

A fitted model is not the end of the analysis. It is the beginning of the model check.

The previous chapters built multiple linear regression from the design-matrix point of view:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad \hat{\mathbf{y}} = \mathbf{H}\mathbf{y}, \quad \mathbf{e} = \mathbf{y} - \hat{\mathbf{y}}. \quad (12.1)$$

The residual vector $\mathbf{e}$ is the part of the observed response vector that the fitted model did not explain. Residual analysis asks:

Does the leftover pattern look like random noise, or does it look like structure the model failed to capture?

This chapter is about using residuals as a diagnostic tool and then deciding how to repair a model when the diagnostics look bad. The three practical repair moves emphasized in the source slides are:

1. transform the predictors and/or the response,
2. add interaction terms,
3. admit that the current linear-model frame is not adequate and move to a different modeling strategy.

The most important habit is to avoid treating a single hypothesis test, a single R^2 , or a single residual plot as the whole story. Residual analysis is evidence. Test MSE, cross-validation, domain knowledge, and interpretability are also evidence.

As in Chapters 7, 9, and 11, we use $\mathbf{y}$ for the observed response vector from the fitted data set. Chapter 10 used $\mathbf{Y}$ when the response vector was being treated as random. Here the emphasis is back on the realized fit and its residuals.

Math Foundations Connection. Residual analysis is geometric, but the supporting Math Foundations ideas are more basic: norms, orthogonality, matrix-vector multiplication, rank, column-space/residual-space geometry, and least-squares minimization. These ideas are covered in the Math Foundations textbook, Chapters 5, 7, 10, 11, and 12. The projection language used in this chapter

is a Data Analysis synthesis built from those underlying ideas. Model repair changes the design columns and therefore changes which patterns can be represented by the fitted part of the model.

12.2 Recall from Chapters 7 and 10: The Residual Vector

Chapters 7 and 10 already developed the linear-algebra machinery behind residuals. We recall only the pieces needed for diagnostics.

For a full-rank design matrix $\mathbf{X} \in \mathbb{R}^{n \times (p+1)}$, ordinary least squares gives

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}, \quad \hat{\mathbf{y}} = \mathbf{H} \mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T. \quad (12.2)$$

The residual vector is therefore

$$\boxed{\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{y}.} \quad (12.3)$$

If the linear truth-plus-error model is appropriate, then

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}. \quad (12.4)$$

Since $\mathbf{H}\mathbf{X} = \mathbf{X}$, the residuals can be written as

$$\boxed{\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\epsilon}.} \quad (12.5)$$

This identity is the key diagnostic reminder: residuals are not the true errors. They are the parts of the errors left over after projecting onto the fitted model space.

The normal equations also give

$$\mathbf{X}^T \mathbf{e} = \mathbf{0}. \quad (12.6)$$

Thus residuals are orthogonal to every column of the design matrix. If the model includes an intercept column $\mathbf{1}$, then

$$\mathbf{1}^T \mathbf{e} = \sum_{i=1}^n e_i = 0. \quad (12.7)$$

This is an algebraic fact, not a diagnostic victory. The residuals can sum to zero and still show curvature, changing spread, time dependence, or outliers. The modeling assumption we care about is closer to

$$\mathbb{E}[\epsilon_i \mid \mathbf{x}_i] = 0, \quad (12.8)$$

meaning that the unobserved errors are centered correctly at each predictor setting. Residual plots give practical evidence for or against that claim.

12.3 What Residuals Are Trying to Tell Us

The source slides summarize the residual-checking logic as follows:

Residual behavior	Suggests about the errors	Common diagnostic view
Mean-zero pattern for each x	$\mathbb{E}[\epsilon \mid X = x] = 0$ may be plausible	residuals versus fitted values or predictors
Approximately constant spread	$\text{Var}(\epsilon \mid X = x) = \sigma^2$ may be plausible	residuals versus fitted values; scale-location plot
Approximately normal shape	$\epsilon \sim N(0, \sigma^2)$ may be plausible	histogram or normal Q-Q plot
No sequential pattern	uncorrelated errors may be plausible	residuals versus time/order

The phrase “suggests” is important. Residual plots do not prove assumptions. They help us identify visible violations.

12.4 Residuals Under the Big Three Regression Assumptions

Residual diagnostics connect directly to the three-step assumption ladder from earlier regression chapters. This section gives the diagnostic consequences, rather than re-deriving the full residual theory.

12.4.1 Assumption 1: Mean-zero errors

If $\mathbb{E}[\epsilon \mid \mathbf{X}] = \mathbf{0}$, then the model is centered correctly and residuals should not show systematic mean structure. A residual plot with no visible pattern around zero supports this assumption. A curved residual pattern suggests that the mean model is missing structure, such as a transformed predictor, a polynomial term, or an interaction.

12.4.2 Assumption 2: Constant variance and uncorrelated errors

If $\text{Var}(\epsilon \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$, then the usual least-squares standard-error formulas are supported. Chapter 10 showed that residuals then satisfy

$$\text{Var}(\mathbf{e} \mid \mathbf{X}) = \sigma^2(\mathbf{I}_n - \mathbf{H}), \quad \text{Var}(e_i \mid \mathbf{X}) = \sigma^2(1 - h_{ii}). \quad (12.9)$$

The important practical point is that raw residuals are affected by leverage. Changing spread in residual plots suggests heteroskedasticity, while waves or runs over time/order suggest correlated errors.

12.4.3 Assumption 3: Gaussian errors

If $\epsilon \mid \mathbf{X} \sim N(\mathbf{0}, \sigma^2 \mathbf{I}_n)$, then the usual finite-sample t - and F -based inference is exact under the model. Normal Q-Q plots and residual histograms are used to look for strong skewness, heavy tails, or outliers. For prediction, mild nonnormality may be less damaging than nonlinearity, heteroskedasticity, or dependence.

12.5 Residual Scaling: Raw, Standardized, and Studentized

Raw residuals are useful for seeing patterns, but they do not all have the same variance because leverage changes residual variability. Standardized and internally studentized residuals rescale residuals so unusually large residuals are easier to spot; externally studentized residuals use a leave-one-out estimate of σ . The full reference text gives the detailed formulas.

12.6 Core Diagnostic Plots

12.6.1 Residuals versus fitted values

This is the primary diagnostic plot for linear regression. The horizontal axis is $\hat{y}_i$, and the vertical axis is e_i . A good residual-versus-fitted plot looks like an unstructured cloud centered around zero.

Common patterns include:

Pattern	Likely issue	Candidate repair
Curved shape	missing nonlinear mean structure	polynomial term, log term, spline, or other transformed predictor
Fan shape	nonconstant variance	transform response, weighted least squares, robust standard errors
Separate bands by group	missing categorical effect or interaction	add group indicator or interaction
Large isolated points	outliers or data errors	inspect data, sensitivity analysis, robust method

12.6.2 Residuals versus predictors

A residual-versus-fitted plot may hide which predictor is causing the problem. Plotting residuals against individual predictors can reveal a missed nonlinear effect in one variable.

For example, if residuals plotted against x_1 show a curve, then a candidate repair is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_1^2 + \cdots + \epsilon. \quad (12.10)$$

If residuals plotted against x_2 show increasing spread, then a transformation or weighting strategy may be needed.

12.6.3 Normal Q-Q plot

A normal Q-Q plot compares standardized residual quantiles to normal quantiles. If the residuals are approximately normal, the points should lie roughly along a straight line.

Common departures include:

1. heavy tails: ends bend away from the line,

2. skewness: one tail departs more than the other,
3. isolated outliers: one or a few points far from the line.

12.6.4 Scale-location plot

A scale-location plot often uses

$$\sqrt{|r_i|} \tag{12.11}$$

against fitted values. It is designed to make changing variance easier to see. A rising trend suggests that the residual spread increases with the fitted value.

12.6.5 Residuals versus time or collection order

If the observations have a natural order, plot residuals against time, mission phase, batch number, route order, or collection order. A visible wave, trend, or clustering pattern can indicate correlated errors or missing time-varying predictors.

12.7 Problem 1: Nonlinearity

A linear model assumes that the conditional mean is linear in the columns of the design matrix:

$$\mathbb{E}[Y \mid \mathbf{x}] = \mathbf{x}^T \boldsymbol{\beta}. \tag{12.12}$$

If the true mean function is curved and the design matrix has only straight-line columns, residuals will show structure.

Diagnostic sign

A residual plot with a curve is the classic sign. For example, a straight-line fit to a parabola will often leave residuals that are negative-positive-negative across the predictor range.

Repairs

Candidate repairs include:

1. adding polynomial terms such as x^2 or x^3 ,
2. using transformations such as $\log x$, $\sqrt{x}$, or $1/x$,
3. using sine/cosine terms for cyclic predictors,
4. using splines or generalized additive models when a simple transformation is not enough,
5. moving to a different model class if the response type requires it.

Model repair as new columns

If the original model is

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \quad (12.13)$$

then adding a quadratic predictor changes the design matrix to

$$\mathbf{Z} = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}. \quad (12.14)$$

The repaired model is

$$\mathbf{y} = \mathbf{Z}\boldsymbol{\theta} + \boldsymbol{\epsilon}. \quad (12.15)$$

So model repair is still a projection problem; we changed the subspace.

12.8 Problem 2: Nonconstant Variance

The homoskedastic model assumes

$$\text{Var}(\epsilon_i \mid \mathbf{x}_i) = \sigma^2 \quad (12.16)$$

for every observation. If residual spread changes with fitted values or predictor values, then the constant-variance assumption is questionable.

Diagnostic sign

A common pattern is a funnel shape: residuals are tightly clustered for small fitted values and widely spread for large fitted values, or vice versa.

Why it matters

Nonconstant variance can make standard errors, confidence intervals, and hypothesis tests unreliable. It may not destroy point predictions, but it can damage uncertainty quantification.

Repairs

Candidate repairs include:

1. transform the response, especially $\log Y$ or $\sqrt{Y}$,
2. use weighted least squares when the variance pattern is understood,
3. use heteroskedasticity-robust standard errors for inference,
4. use a model family designed for the response type, such as logistic or Poisson regression.

A log response model has the form

$$\log(Y_i) = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (12.17)$$

which often helps when variability grows multiplicatively with the mean.

A square-root response model has the form

$$\sqrt{Y_i} = \beta_0 + \beta_1 x_i + \epsilon_i, \quad (12.18)$$

which can help for nonnegative count-like responses.

12.9 Problem 3: Nonnormal Residuals

The Gaussian noise model assumes

$$\epsilon_i \sim N(0, \sigma^2). \quad (12.19)$$

A histogram or Q-Q plot of residuals can reveal skewness, heavy tails, or outliers.

Why it matters

Normality is most important for exact small-sample inference. If the sample is large, coefficient estimates may still behave well under weaker conditions, but heavy tails and outliers can still destabilize the fit.

Repairs

Candidate repairs include:

1. transform the response,
2. inspect and correct data-entry issues,
3. use robust regression or bootstrap uncertainty intervals,
4. use a model family more appropriate to the response distribution.

12.10 Problem 4: Outliers and Influence

An outlier is an observation with a response value far from what the model predicts. A high-leverage point is an observation with an unusual predictor vector. A point can be one, both, or neither.

The leverage of observation i is

$$h_{ii} = \mathbf{x}_i^T (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{x}_i. \quad (12.20)$$

Here $\mathbf{x}_i^T$ is the full design row for observation i , including the intercept and any transformed or interaction columns. High leverage is not automatically bad. It means the observation is far out in predictor space and can have a strong effect on the fitted model.

A common influence measure is Cook's distance:

$$D_i = \frac{e_i^2}{(p+1)\hat{\sigma}^2} \frac{h_{ii}}{(1-h_{ii})^2}. \quad (12.21)$$

Large Cook's distance suggests that deleting observation i would noticeably change the fitted model.

Repair logic. Do not delete a point merely because it is inconvenient. First ask:

1. Was the observation recorded correctly?
2. Is it inside the intended population of analysis?
3. Does it represent an operationally important rare case?
4. Does the conclusion change if the point is included versus excluded?

A defensible analysis often reports a sensitivity check.

12.11 Problem 5: Correlated Errors

Least squares formulas for standard errors often assume uncorrelated errors:

$$\text{Cov}(\epsilon_i, \epsilon_j \mid \mathbf{X}) = 0 \quad \text{for } i \neq j. \quad (12.22)$$

This assumption is especially vulnerable when observations are collected over time, across space, within units, or by repeated measurements on the same platform or person.

Diagnostic sign

Residuals plotted over time or collection order may show runs, waves, or clusters. Adjacent residuals may have similar signs or magnitudes.

Repairs

Candidate repairs include:

1. include time, batch, route, or group predictors,
2. include lagged predictors or lagged responses when appropriate,
3. use generalized least squares or time-series models,
4. use clustered or block-robust standard errors,
5. redesign the data collection plan when possible.

12.12 Problem 6: Missing Interactions

An additive model assumes that the effect of one predictor does not depend on another predictor. Interactions repair that limitation.

The additive two-predictor model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon. \quad (12.23)$$

The interaction model is

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 + \epsilon. \quad (12.24)$$

Equivalently,

$$Y = \beta_0 + (\beta_1 + \beta_3 x_2)x_1 + \beta_2 x_2 + \epsilon. \quad (12.25)$$

The slope in x_1 is now allowed to depend on x_2 .

Diagnostic signs

Possible signs of a missing interaction include:

1. residuals have different patterns in different groups,
2. one predictor appears helpful only at certain levels of another predictor,
3. residuals plotted by color or symbol for a second variable reveal structure,
4. domain knowledge says the variables should combine synergistically.

12.13 A Model Repair Menu

When residuals look bad, the repair should be tied to the defect.

Residual symptom	Likely interpretation	First repair candidates
Curved residual trend	wrong mean shape	transform predictor, polynomial, spline
Fan-shaped spread	heteroskedasticity	transform response, WLS, robust SEs
Heavy tails in Q-Q plot	outliers or nonnormal errors	inspect data, robust method, transformation
Runs over time/order	correlated errors	time terms, lags, GLS, clustered SEs
Different patterns by group	missing categorical effect or interaction	add indicators or interactions
Large influential point	unusual observation dominates fit	audit, sensitivity analysis, robust method
No repair makes sense	model class mismatch	change model family or data strategy

12.14 Transformation Repairs

The source slides compare several repair candidates: a linear fit, a polynomial predictor transformation, a square-root transformation, and a logarithmic-linear transformation. The point is not that one transformation is always best. The point is to use residual behavior, EDA intuition, domain knowledge, and test MSE together. A transformation is a modeling decision, not a cosmetic fix.

12.15 Which One Wins?

The source slides emphasize that model repair includes some qualitative judgment. A defensible answer combines several checks:

1. Do residual plots look less patterned?
2. Does test MSE or cross-validated error improve on the operational prediction scale?
3. Is the transformation plausible for the domain?
4. Does the repair preserve interpretability?
5. Did the repair create new problems, such as extrapolation risk or rank deficiency?
6. Are nested-model tests, adjusted R^2 , AIC, or BIC consistent with the conclusion?

Nested-model comparison

If a reduced model is nested inside a full model, use the familiar comparison statistic:

$$F = \frac{(RSS_R - RSS_F)/(df_R - df_F)}{RSS_F/df_F}. \quad (12.26)$$

For example, the linear model

$$Y = \beta_0 + \beta_1 x + \epsilon \quad (12.27)$$

is nested inside the cubic model

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon. \quad (12.28)$$

The test asks whether the added polynomial columns explain enough additional variation to justify the added degrees of freedom.

Prediction-focused comparison

For prediction, validation should dominate training fit. The model with the smallest training RSS is often the most flexible model, not necessarily the best generalizing model. Use a train/test split or cross-validation to estimate test error.

Inference-focused comparison

For inference, residual repair should support assumptions needed for standard errors, intervals, and tests. A model with slightly worse prediction but clearer interpretation and better-behaved residuals may be the better briefing model.

12.16 Admit Defeat: When Repair Means Changing the Model Class

Sometimes bad residuals are not a minor defect. They are evidence that ordinary least squares is the wrong tool.

Examples:

Situation	Why OLS may struggle	Better direction
Binary response	predictions can fall outside $[0, 1]$	logistic regression
Count response	variance often grows with mean	Poisson or negative binomial model
Positive skewed response	additive normal errors may be implausible	log transform or Gamma GLM
Repeated measures	errors are correlated within unit	mixed model or clustered SEs
Time series	residuals may be autocorrelated	time-series regression or GLS
Strong nonlinear boundary	linear mean is structurally wrong	nonlinear model, tree, GAM, or ML method

Admitting defeat is not failure. It is a modeling conclusion: the residuals are telling us the current model family is missing something important.

12.17 Worked Example: Residual Pattern to Repair Decision

A useful residual-repair workflow is: identify the visible pattern, connect it to a violated assumption or missing structure, propose one targeted repair, refit, and compare both residual behavior and prediction performance. For example, a curved residual pattern suggests missing nonlinear mean structure; a reasonable repair is to add a transformed predictor column, then check whether the residual plot and test MSE improve.

12.18 Residual Analysis Workflow

For the Core Reader, use this compact workflow:

1. Fit the model and compute residuals.
2. Plot residuals against fitted values and key predictors.
3. Check for curvature, changing spread, outliers, influence, nonnormal shape, or time/order patterns.
4. Connect each pattern to the Big Three regression assumptions or to missing model structure.
5. Try one defensible repair at a time: transform variables, add interactions, address data issues, or change model class.
6. Compare repaired models using residual behavior, test MSE or cross-validation, interpretability, and domain knowledge.

12.19 Chapter Summary

A fitted model is not the end of the analysis. Residual analysis asks whether the leftover pattern looks like random noise or structure the model failed to capture. Residuals are computed from the fitted model; they are not the same as the true errors. With an intercept, residuals sum to zero by construction, so a mean-zero residual average is not enough to validate the model.

Residual plots help diagnose violations of the Big Three regression assumptions and missing model structure. Curvature suggests a poor mean model, fan shapes suggest nonconstant variance, Q–Q plot problems suggest nonnormal residuals, time patterns suggest dependence, and isolated high-leverage points may be influential.

Model repair should be deliberate. Transform predictors or the response, add interactions, address data-quality problems, or admit that the current linear-model frame is inadequate. Do not rely on a single hypothesis test, a single R^2 , or a single residual plot in isolation.

Can you do this? You should be able to explain what a residual plot is checking, connect patterns to assumptions, propose a targeted repair, compare repair candidates, and brief the limitations of the final model. The full reference text contains the common-mistakes list, operational briefing checklist, complete Math Foundations source map, and reusable study template.